FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), I want to use a centralized Sales force-based platform to record, manage, and monitor all farm-related activities in an organized manner. As a farmer, farm manager, or system administrator, I want to maintain information about farms, crops, farmers, buyers, products, and agricultural activities securely.

I want to enter and update farm details, crop information, production records, buyer information, and sales transactions through a simple and user-friendly interface. The system should provide proper validation rules to reduce incorrect data entry and maintain accurate records.

I also want to view real-time reports and dashboards to monitor farm productivity, crop performance, sales, and other important activities. This will help users analyze farm operations and make informed decisions.

Additionally, the system should provide role-based access and secure data storage so that sensitive information is protected. The system will improve collaboration between farmers, managers, buyers, and administrators while reducing manual paperwork, minimizing errors, and improving overall farm management efficiency.

1. INTRODUCTION

1.1 Project Overview:

The **Farm Management System (FMS)** is a Salesforce-based application developed to digitize and simplify various agricultural and farm management activities. The system provides a centralized platform for managing information related to farms, crops, farmers, buyers, products, and sales transactions.

The application helps users maintain accurate records and access important agricultural information easily. It also provides reports and dashboards that help farm managers monitor crop production, farm activities, sales, and overall performance.

1.2 Purpose:

The main purpose of the Farm Management System is to provide an efficient and secure platform for managing farm-related information and activities.

The system is designed to:

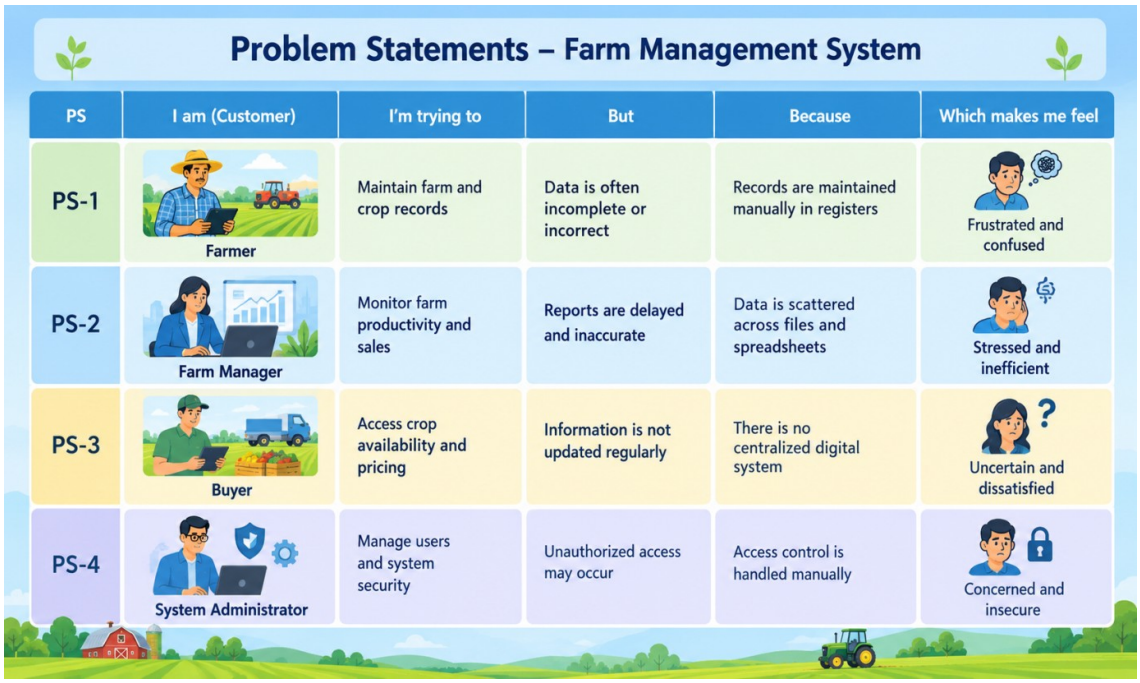
- Maintain farm and farmer information.
- Manage crop and production details.
- Store buyer and product information.
- Manage sales and transaction records.
- Reduce manual data entry and paperwork.
- Improve data accuracy and consistency.
- Provide reports and dashboards for analysis.
- Support better decision-making.
- Provide secure and role-based access to information.
- Improve overall farm management efficiency.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are maintained manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and spreadsheets	Stressed and inefficient

PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Administrator	Manage users and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure



2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Agricultural Workers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need simple and reliable farm data.
- **Hear:** Advice from agricultural experts.
- **See:** Farm records and crop information.
- **Say & Do:** Manage and update farm activities.
- **Pain:** Errors, delays, and data loss.
- **Gain:** Easy access and better farm management.

2.3 Brainstorming

The brainstorming process was conducted to identify suitable ideas for improving farm management and reducing manual work.

Ideas Identified:

- Centralized farm data management
- Digital crop record maintenance
- Farmer and buyer management
- Sales and production tracking
- Automated reports and dashboards
- Secure user access
- Data validation and error reduction
- Easy monitoring of farm activities
- Real-time access to important information
- Improved communication between users



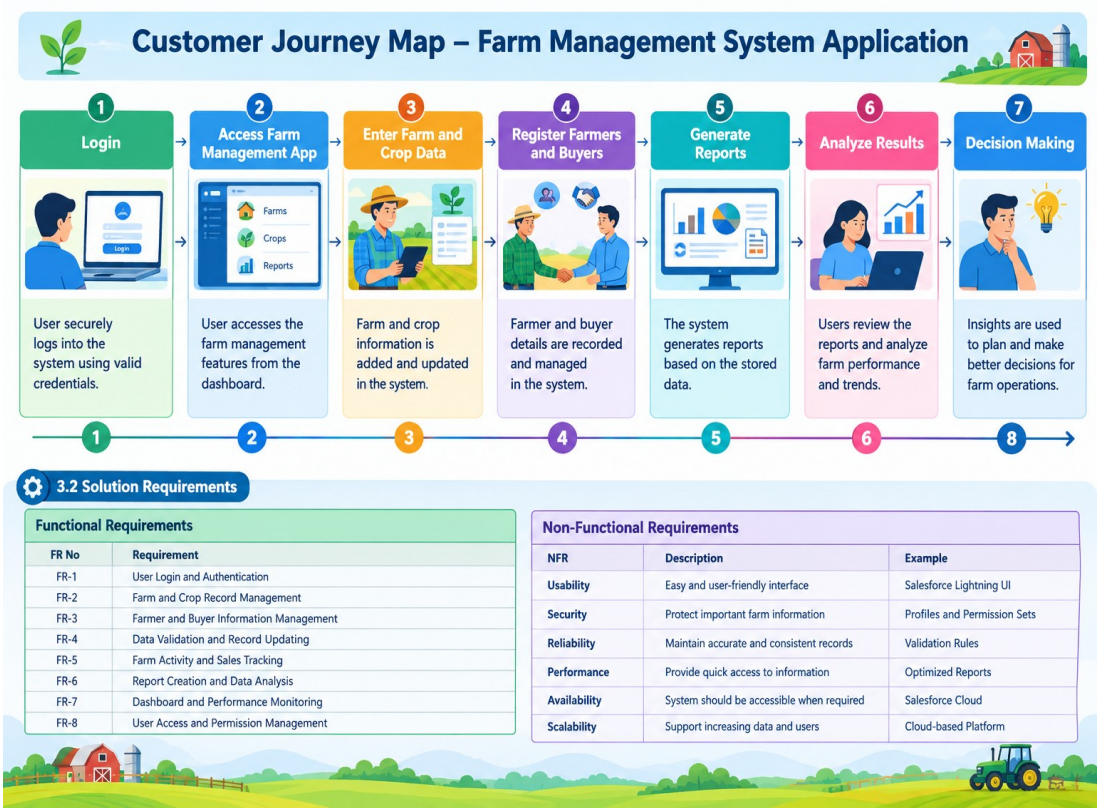
3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. **User Login** – User securely logs into the system.
2. **Open Farm Dashboard** – User accesses the required farm management features.
3. **Add Farm Details** – Farm and cultivation information is entered into the system.
4. **Maintain Crop Records** – Crop details, production data, and related activities are updated.
5. **Manage Farmers and Buyers** – User records and updates farmer and buyer information.

6. **View Reports** – System generates useful reports based on stored data.
7. **Monitor Performance** – Dashboards are used to understand farm activities and sales.
8. **Take Action** – Users use the available information to plan and manage farm operations.



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Login and Authentication
FR-2	Farm and Crop Record Management
FR-3	Farmer and Buyer Information Management
FR-4	Data Validation and Record Updating
FR-5	Farm Activity and Sales Tracking

FR-6	Report Creation and Data Analysis
FR-7	Dashboard and Performance Monitoring

Non-Functional Requirements

NFR	Description	Example
Usability	Simple and user-friendly interface	Sales force Lightning UI
Security	Protect important farm information	Profiles and Permission Sets
Reliability	Maintain accurate and consistent records	Validation Rules
Performance	Provide quick access to information	Optimized Reports
Availability	System should be accessible when required	Sales force Cloud
Scalability	Support increasing data and users	Cloud-based Platform

3.3 Data Flow Diagram

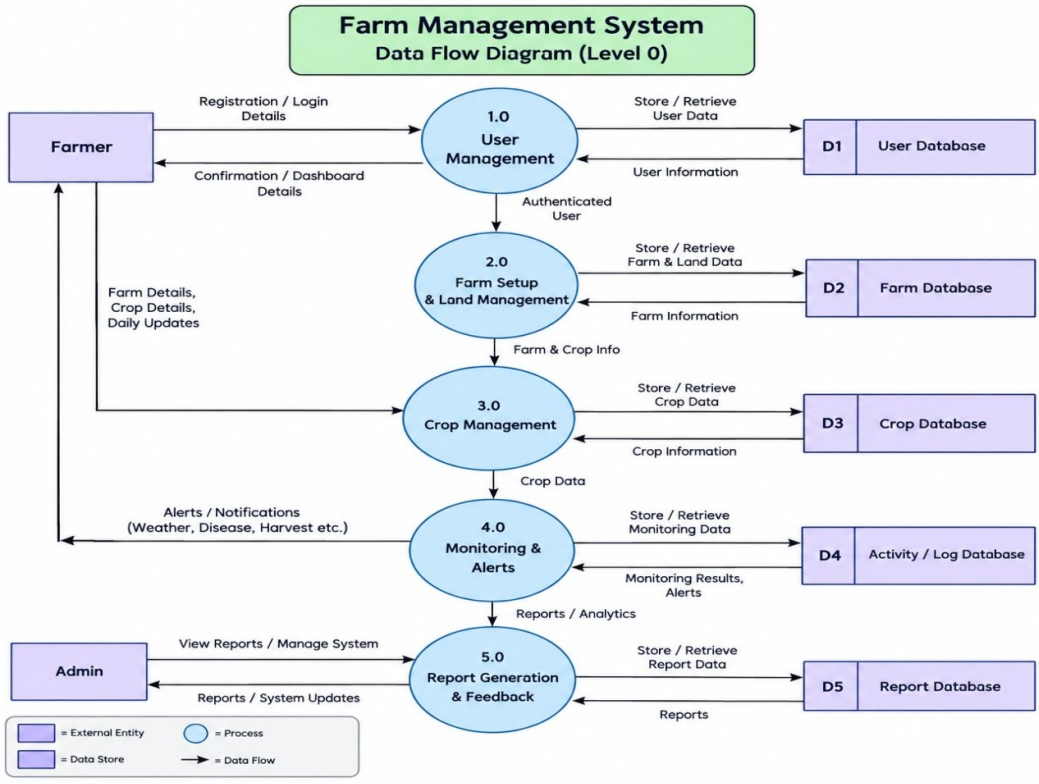
Data Flow:

User → Login → Salesforce Lightning Interface → Farm / Crop / Farmer / Buyer Records → Validation Rules → Data Relationships → Reports & Analytics → Dashboards → User

The system receives information from users through the Salesforce interface. The entered information is stored in appropriate custom objects. Validation rules check the accuracy of the data before it is saved. Relationships connect related records, while reports and dashboards provide useful information for monitoring and analysis.

3.4 Technology Stack

Layer	Technology
User Interface	Salesforce Lightning Experience
Application Platform	Salesforce Platform
Data Storage	Salesforce Cloud Database
Data Management	Custom Objects and Fields
Validation	Validation Rules
Automation	Salesforce Flow
Analytics	Reports and Dashboards



4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The Farm Management System is designed to address the difficulties involved in traditional farm record management. Manual methods make it difficult to organize farm information, update crop details, monitor transactions, and access records quickly.

The proposed system provides a centralized digital platform where farm, crop, farmer, buyer, and sales information can be maintained systematically. Automated validation helps reduce incorrect data, while cloud-based storage provides easier access to records. Reports and dashboards help users monitor farm activities and understand important information.

By integrating these features, the system supports organized data management, faster information retrieval, improved monitoring, and more efficient farm operations



4.2 Proposed Solution

The proposed **Farm Management System (FMS)** is a Salesforce-based application developed to simplify and digitize agricultural management activities. It provides a centralized environment for storing and managing important farm-related information.

The system allows users to maintain details about farms, crops, farmers, buyers, production activities, and sales. Instead of depending on separate notebooks, files, or spreadsheets, users can access the required information through a single platform.

The system also includes validation rules to improve the quality of entered information. Reports and dashboards provide a clear view of farm activities, production, and sales. This helps users monitor operations and obtain useful information whenever required.

The solution can support multiple users and farms through Salesforce's cloud-based environment. Role-based access can also be applied so that users can access information according to their responsibilities.

Overall, the proposed system aims to provide **organized data management, improved accessibility, secure record storage, easier monitoring, and better operational planning**.

4.3 Solution Architecture

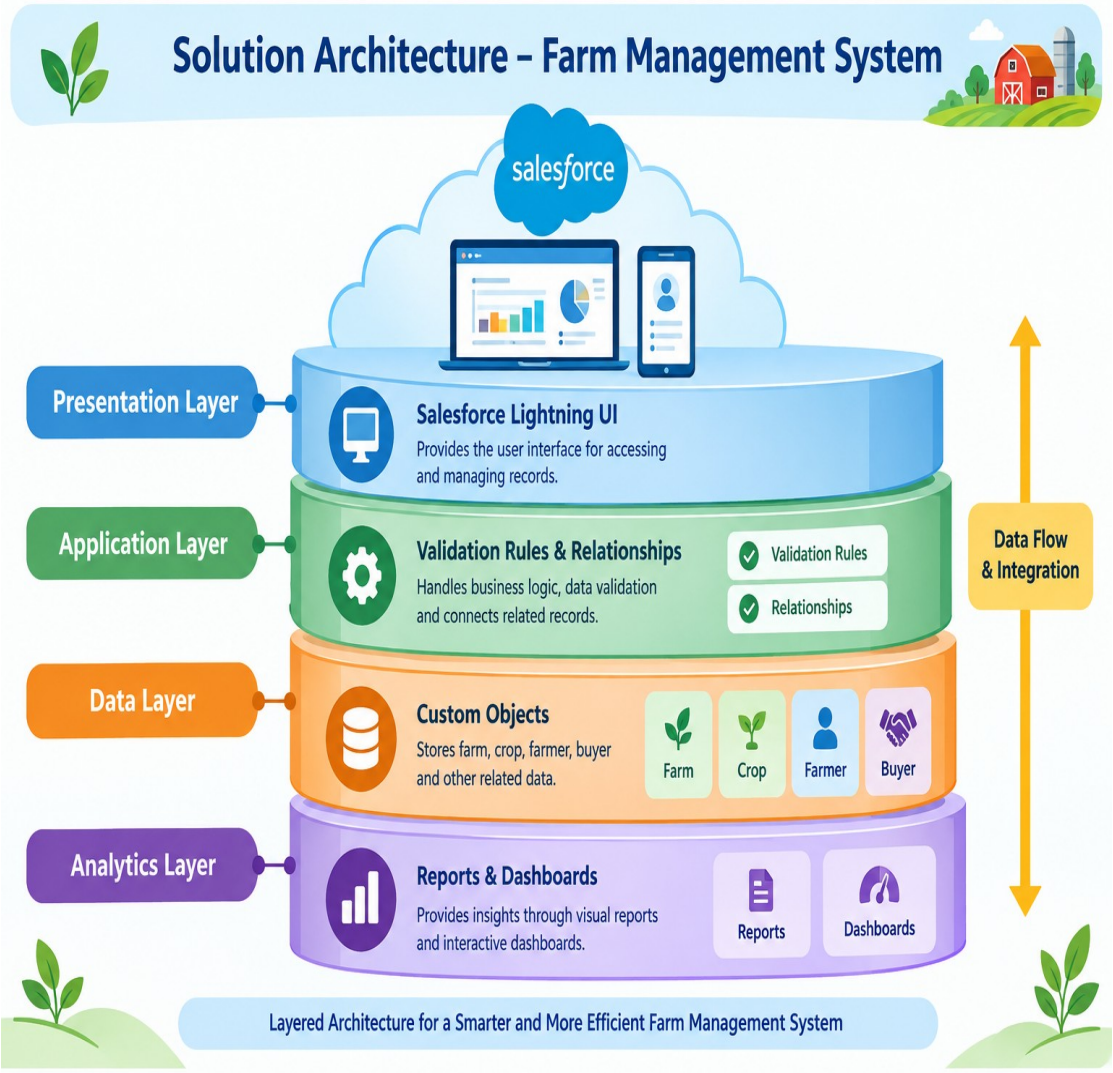
The Farm Management System follows a layered architecture in which each layer performs a specific function.

1. **Presentation Layer:**
   The presentation layer provides the interface through which users interact with the application. Salesforce Lightning provides the user interface for entering records, viewing information, and accessing different features of the system.
2. **Application Layer:**
   The application layer handles the main processing and business rules of the system. Validation rules, record relationships, and automation are used to ensure that information is entered and managed correctly.
3. **Data Layer:**
   The data layer stores the information required by the application. Custom objects and

fields are used to maintain farm, crop, farmer, buyer, production, and sales records in a structured manner.

4. **Analytics Layer:**
   The analytics layer converts stored information into useful reports and dashboards. Users can monitor farm activities, production details, sales information, and other important metrics through visual reports.

- ✓ **Architecture Flow:**
- ✓ **Users → Salesforce Lightning UI → Application Logic → Custom Objects → Reports & Dashboards**



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project planning activities were carried out in a systematic manner to develop the Farm Management System. The major activities included:

- Created and configured the Salesforce development environment.
- Identified the major modules required for farm management.
- Created custom objects for farm-related information.
- Defined fields and established relationships between objects.
- Configured Lightning pages and application navigation.
- Set up user roles and access permissions.
- Implemented validation rules and automation.
- Developed reports for farm and sales information.
- Created dashboards for performance monitoring.
- Performed testing, error correction, and final review.

5.2 Product Backlog, Sprint Schedule and Estimation

Sprint	Functional Requirement / Epic	User Story No.	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Environment Setup	USN-1	As an administrator, I want to configure the Salesforce workspace so that the project environment is ready for development.	3	High	Member 1
Sprint-1	Environment Setup	USN-2	As a developer, I want to configure the Farm Management application and navigation so that users can	3	High	Member 2

			access modules easily.			
Sprint-2	Data Management	USN-3	As an administrator, I want to create Farm, Crop, Farmer, and Buyer objects so that all agricultural information can be maintained systematically.	5	High	Member 2
Sprint-2	Data Management	USN-4	As a user, I want related farm and crop records to be connected so that information can be accessed efficiently.	5	High	Member 3
Sprint-3	Validation & Automation	USN-5	As a staff member, I want validation rules for important fields so that incomplete or incorrect records are avoided.	3	High	Member 4
Sprint-3	Validation & Automation	USN-6	As a manager, I want automated record updates so that repetitive activities can be reduced.	3	Medium	Member 4
Sprint-4	Security	USN-7	As an administrator, I want to assign suitable permissions to different users so that farm information is accessed securely.	5	High	Member 1
Sprint-5	Reporting	USN-8	As a farm manager, I want reports for production, crops, and sales so that farm activities can be reviewed easily.	4	High	Member 3
Sprint-6	Dashboard	USN-9	As a manager, I want visual dashboards for farm performance and sales information so that important trends can be monitored quickly.	4	Medium	Member 4

5.3 Project Tracker, Velocity & Burndown Chart

Sprint	Story Points	Duration	Sprint Start Date	Sprint End Date
Sprint-1	6	5 Days	01 Sep 2026	05 Sep 2026
Sprint-2	10	6 Days	07 Sep 2026	12 Sep 2026
Sprint-3	6	5 Days	14 Sep 2026	18 Sep 2026
Sprint-4	5	5 Days	21 Sep 2026	25 Sep 2026
Sprint-5	4	4 Days	26 Sep 2026	29 Sep 2026
Sprint-6	4	4 Days	30 Sep 2026	03 ct 2026

6. Project Description:

Story Points Completed (as on Planned End Date) 20 Sprint Release Date (Actual) 29 Oct 2022 The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

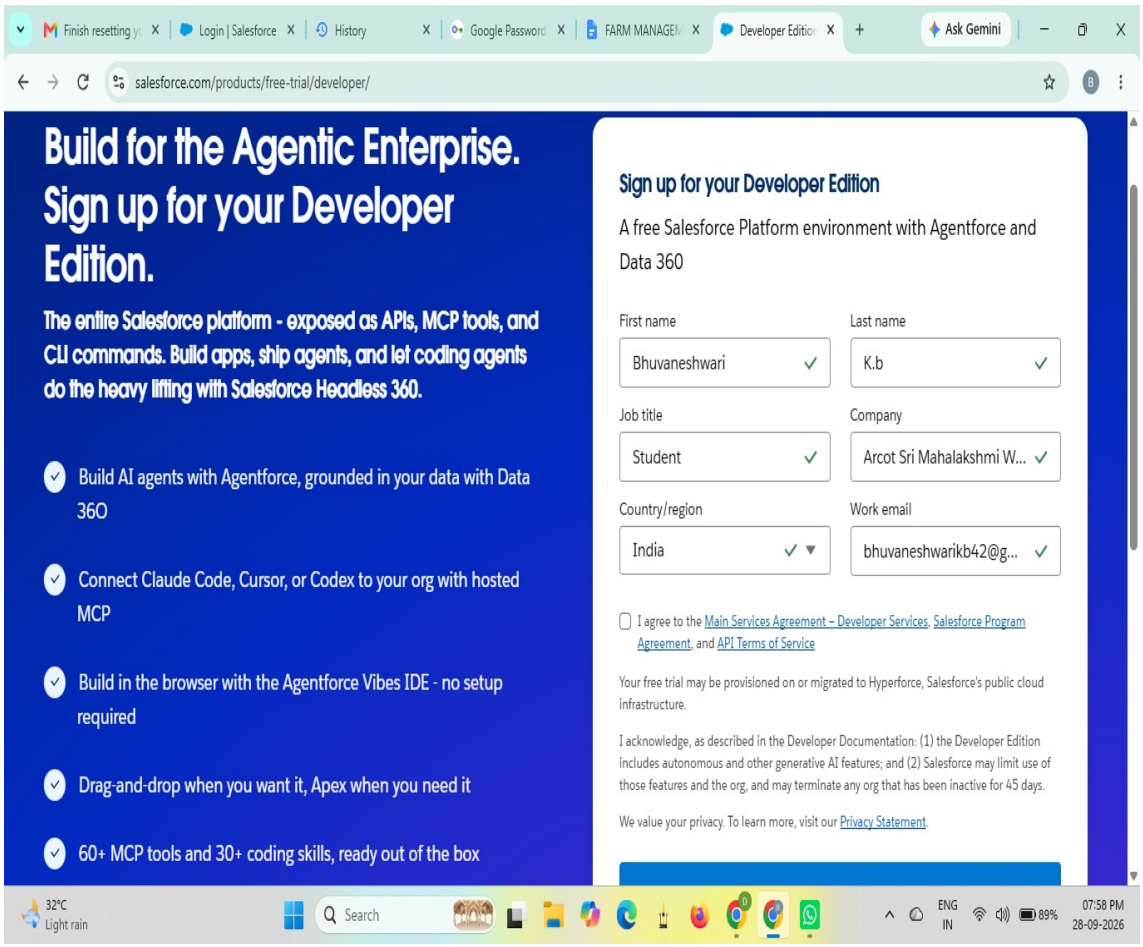
1. Search Developer.salesforce.com/signup

2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.

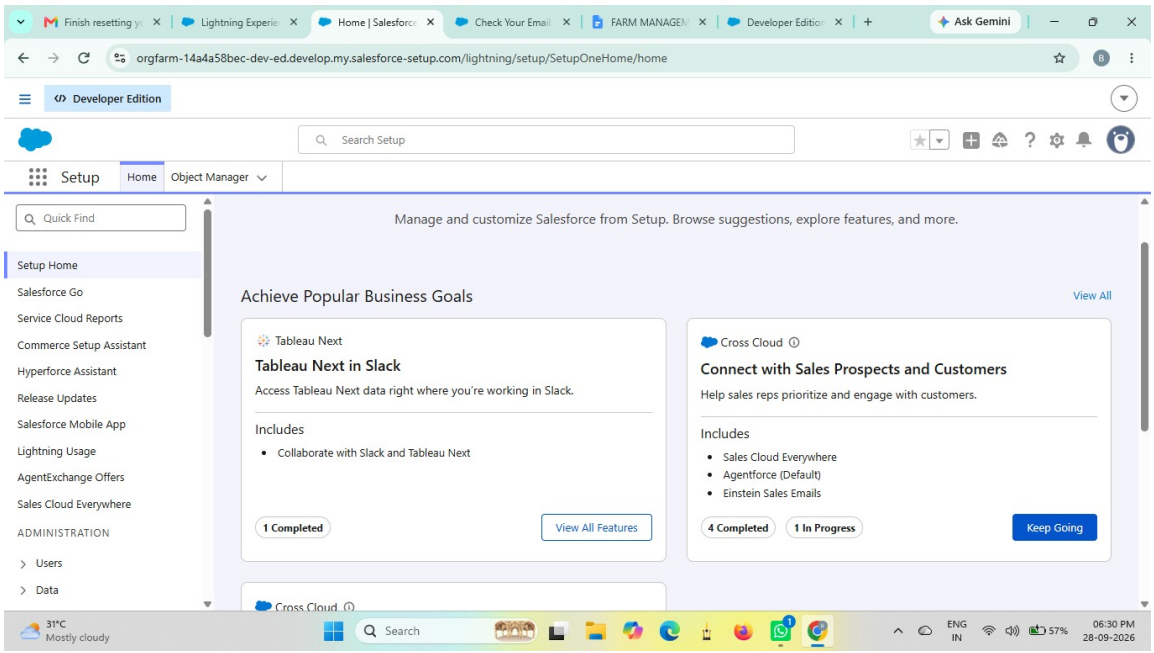
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.

4. Click save.

5. Search login.salesforce.com



6.By using username and password you can into the Salesforce org. The homepage will appear below



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

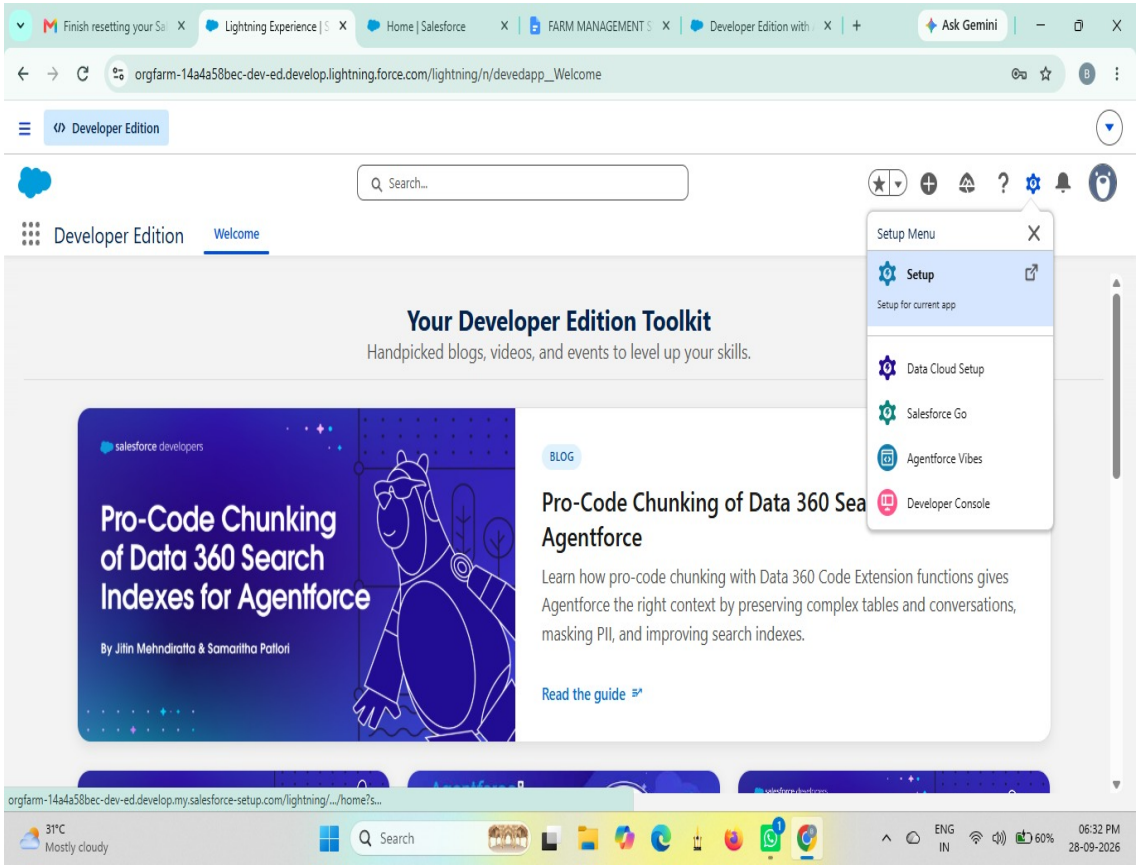
1. Farm
2. Crop
3. Farmer
4. Buyer

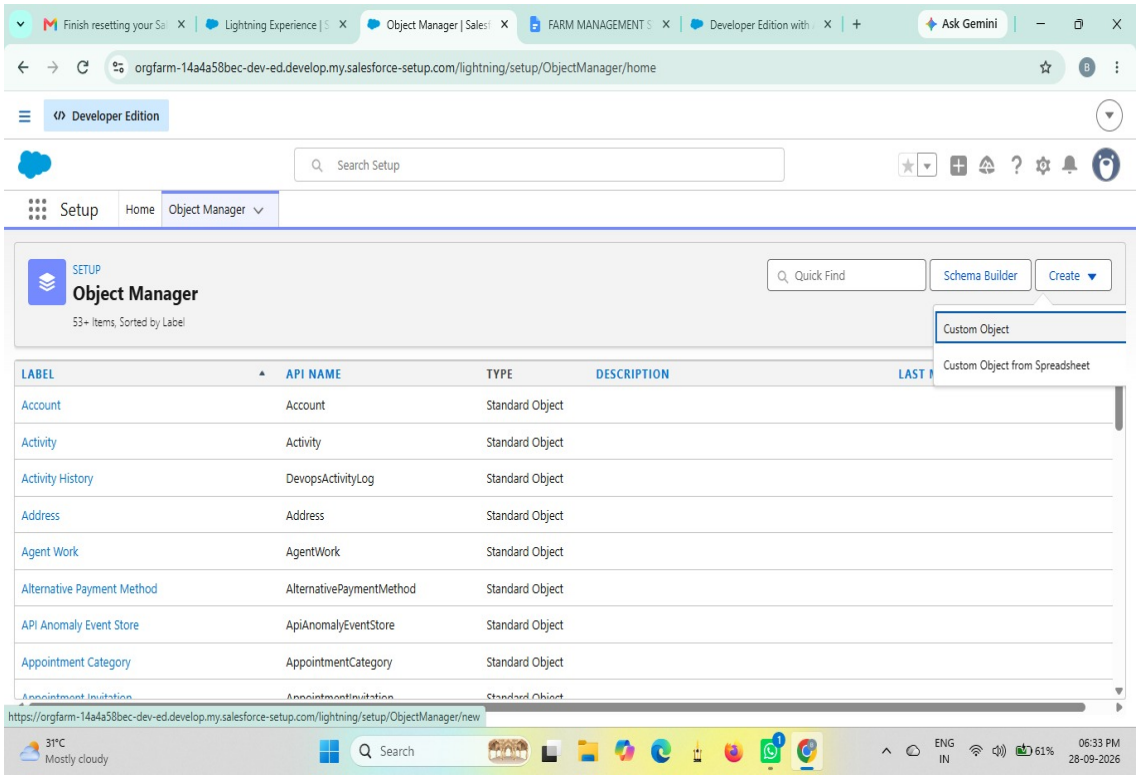
Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.


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Developer Edition
Welcome
Setup Menu
Setup
Setup for current app
Data Cloud Setup
Salesforce Go
Agentforce Vibes
Developer Console
Your Developer Edition Toolkit
Handpicked blogs, videos, and events to level up your skills.
salesforce developers
Pro-Code Chunking of Data 360 Search Indexes for Agentforce
By Jitin Mehndiratta & Samaritha Patlori
BLOG
Pro-Code Chunking of Data 360 Sea
Agentforce
Learn how pro-code chunking with Data 360 Code Extension functions gives Agentforce the right context by preserving complex tables and conversations, masking PII, and improving search indexes.
Read the guide
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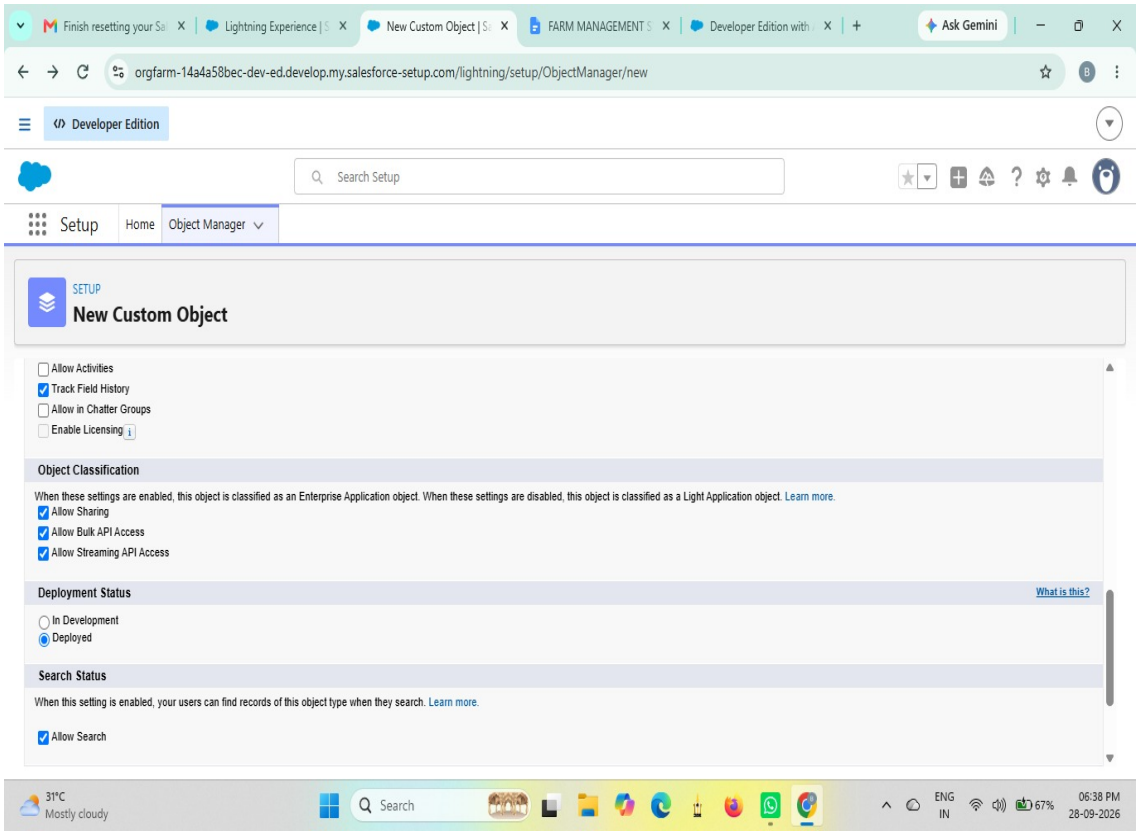

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Search
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3. Fill in the label as " Farm ".

4. Fill in the plural label as " Farms".

5. Record name: Farm Name"

6. Select the data type as "Text".



7. In the Optional Features section, select Allow Reports and Track Field History.

8. In the Deployment Status section, ensure Deployed is selected.

9. In the Search Status section, select Allow Search.

10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:

b. Label: Crop

c. Plural Label: Crops

d. Record Name: Crop Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox

g. Check the Allow Search checkbox

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:

b. Label: Farmer

c. Plural Label: Farmers

d. Record Name: Farmer Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox.

g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:

b. Label: Buyer

c. Plural Label: Buyers

d. Record Name: Buyer Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox.

g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

i. Click Save.

Milestone 3 - Tab: Tab

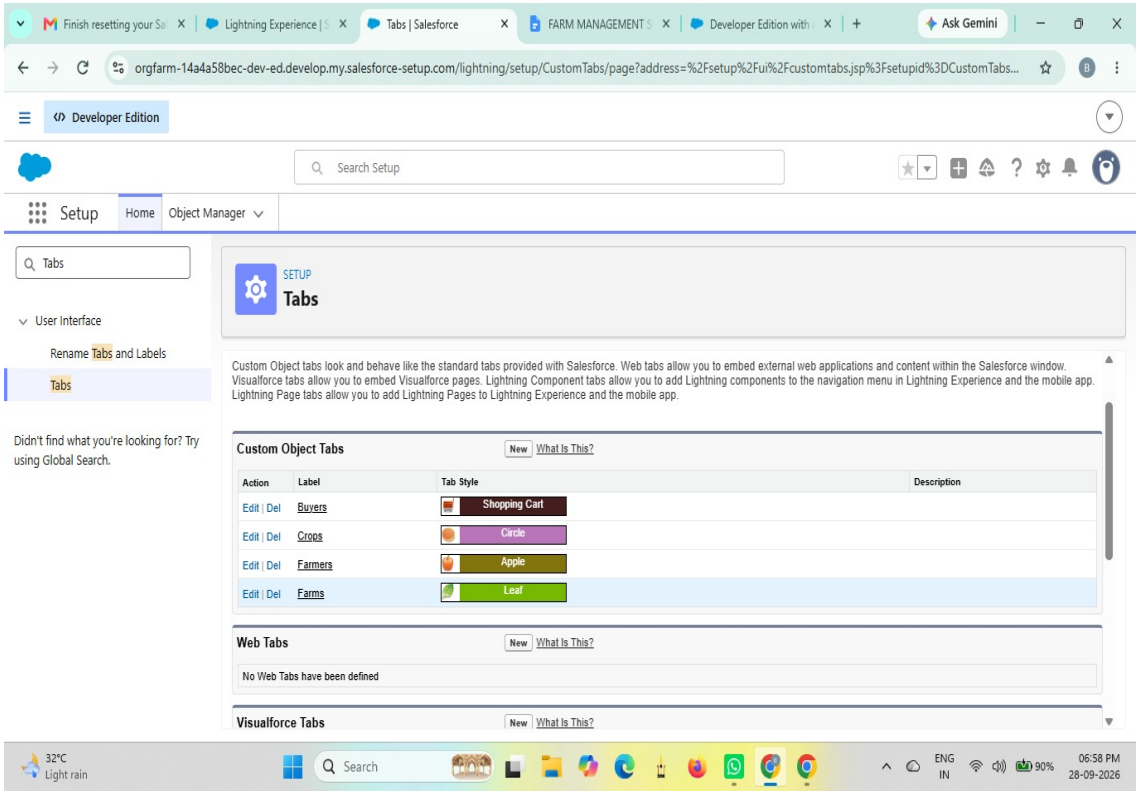
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

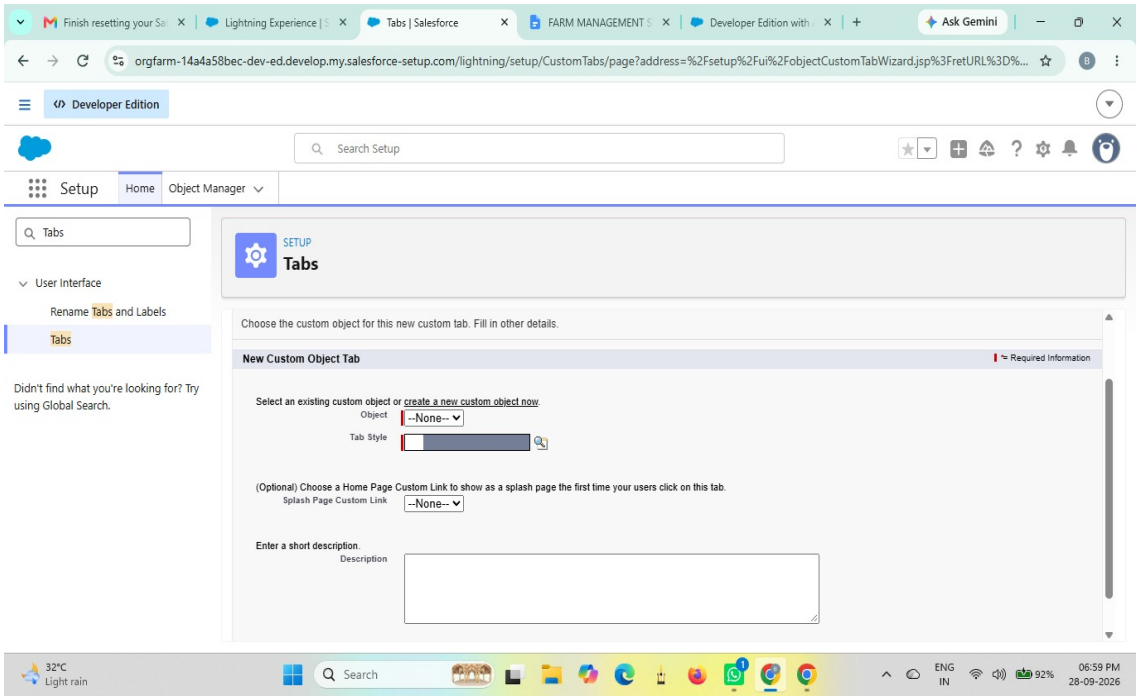
Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2.Under custom object tabs, click New.



3. For Object, select Farm.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Crop.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Farmer.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Buyer

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

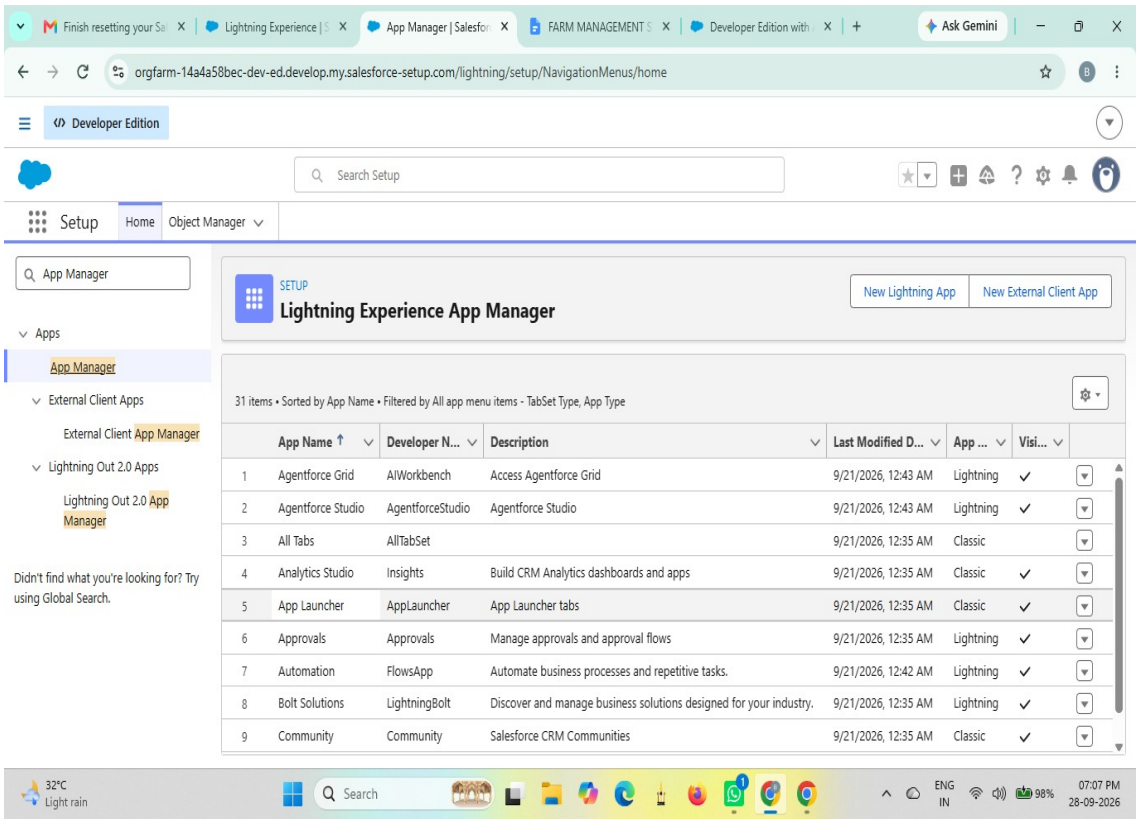
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together asa unit. It has a name,a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



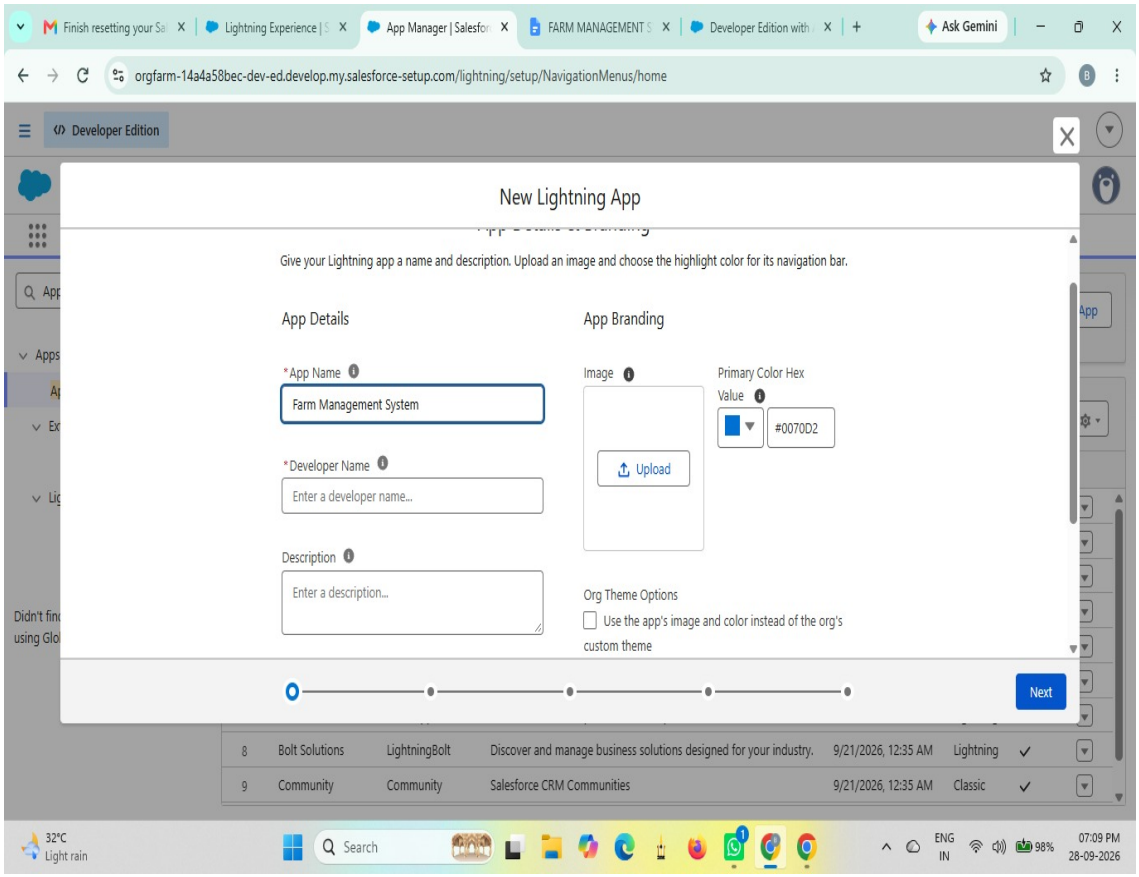
- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop,Farmer,Buyer Reports,and Dashboards and move them to Selected Items. Click Next.

● From Available Profiles, select System Administrator and move it to Selected Profiles.Click Save & Finish.

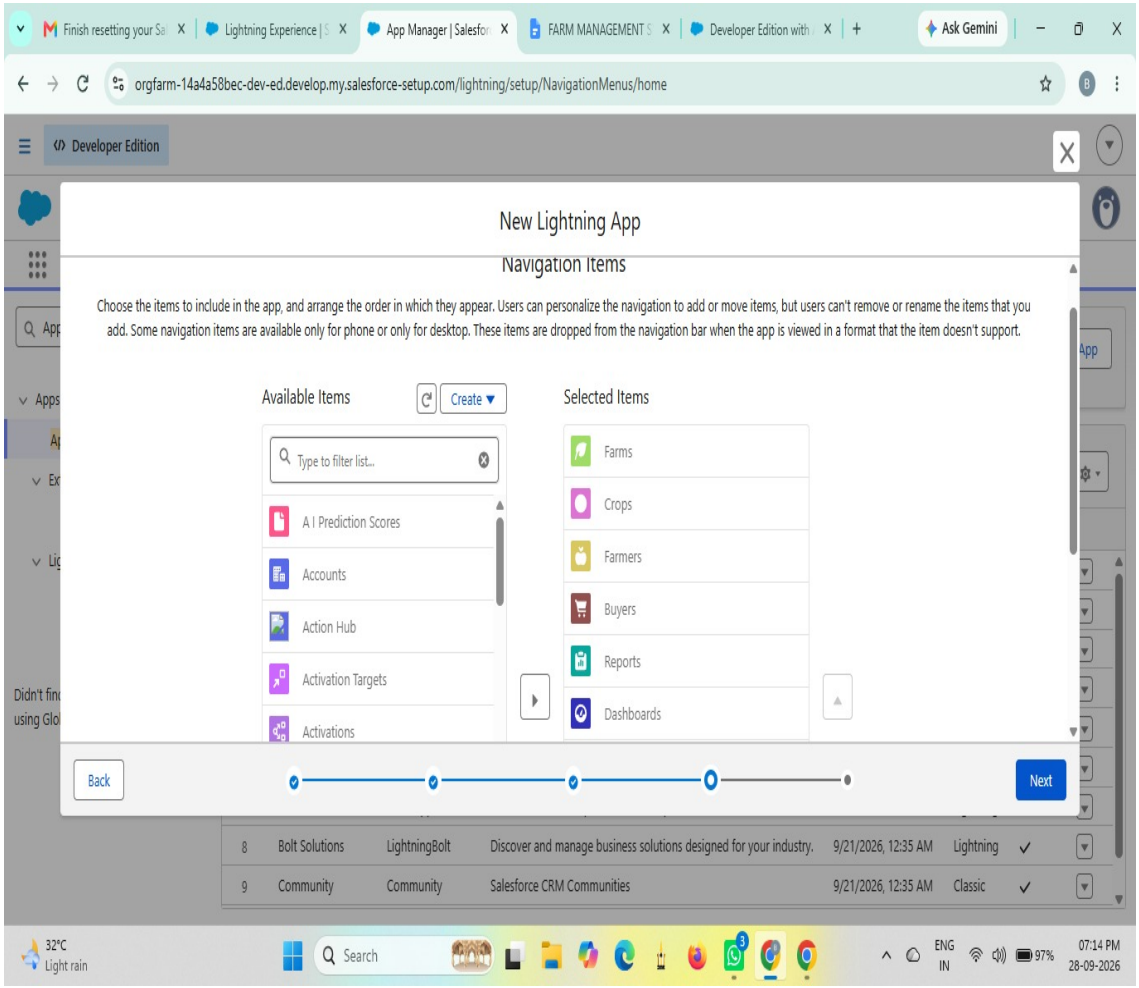
● To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.



2. Navigation menu-Displays the tabs available inside the app.



Milestone 5 - Fields and Relationship:

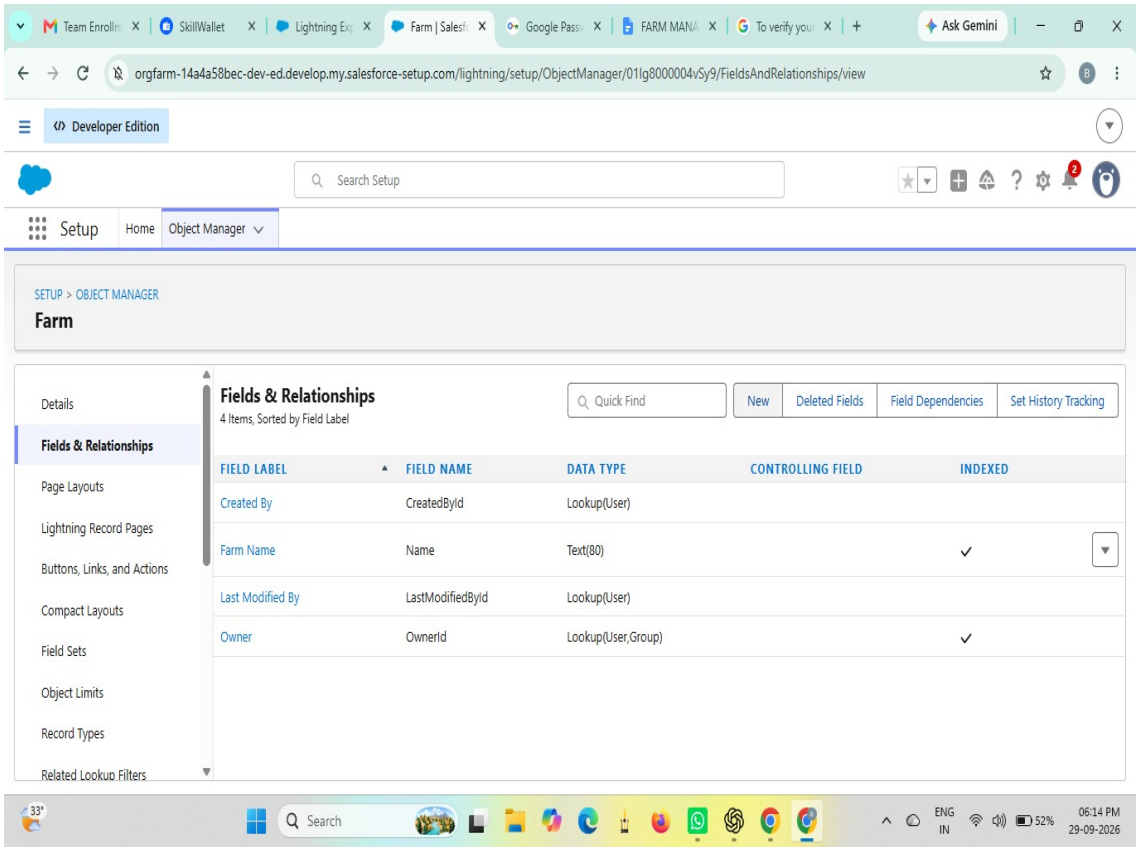
What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

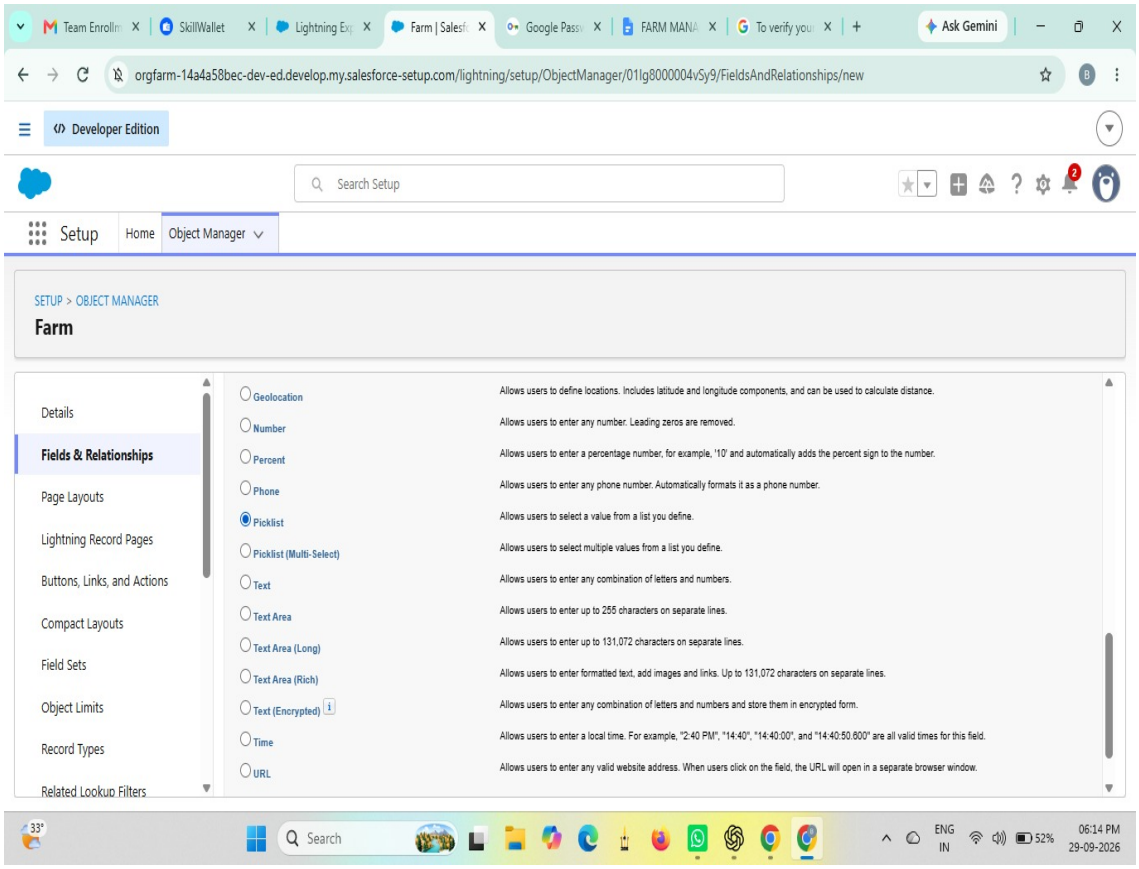
Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New

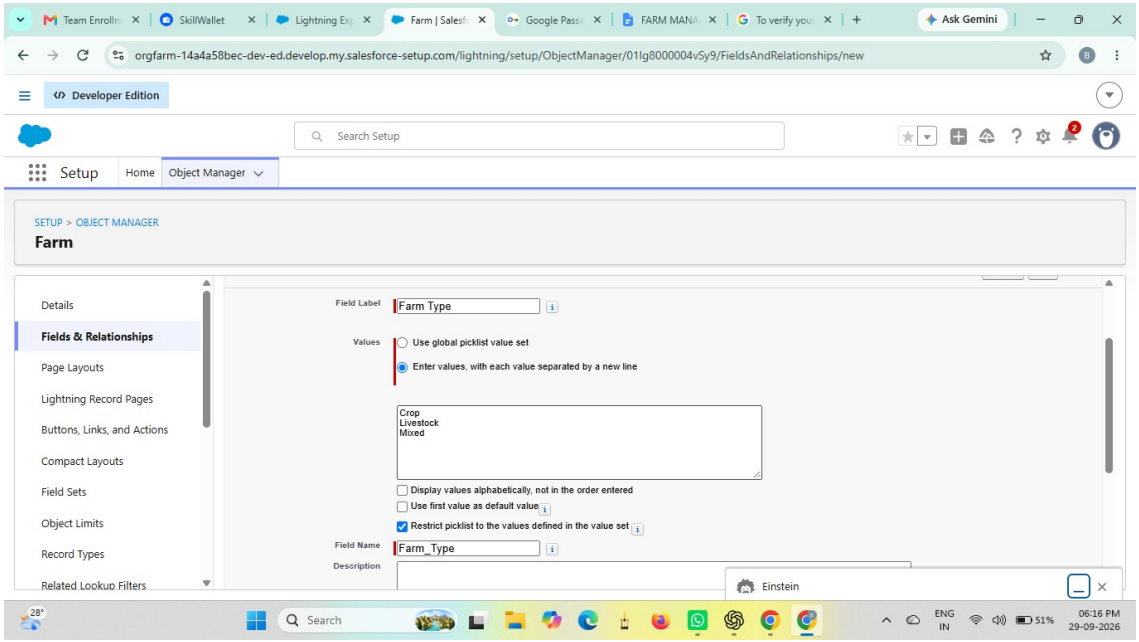


5.Choose Data type as a picklist,Click next



6.Enter Field Label (Farm Type),and Enter Values as (Crop,Livestock,Mixed)

7.Next,Next Click Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)

2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)

3.Choose data type as a text,enter field label(Location)

4.Choose data type as text,enter field label (size)

5.Create a Master Detail relationShip with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save.

2.Select Data type as date ,Enter Field label (Planting Date),then save .

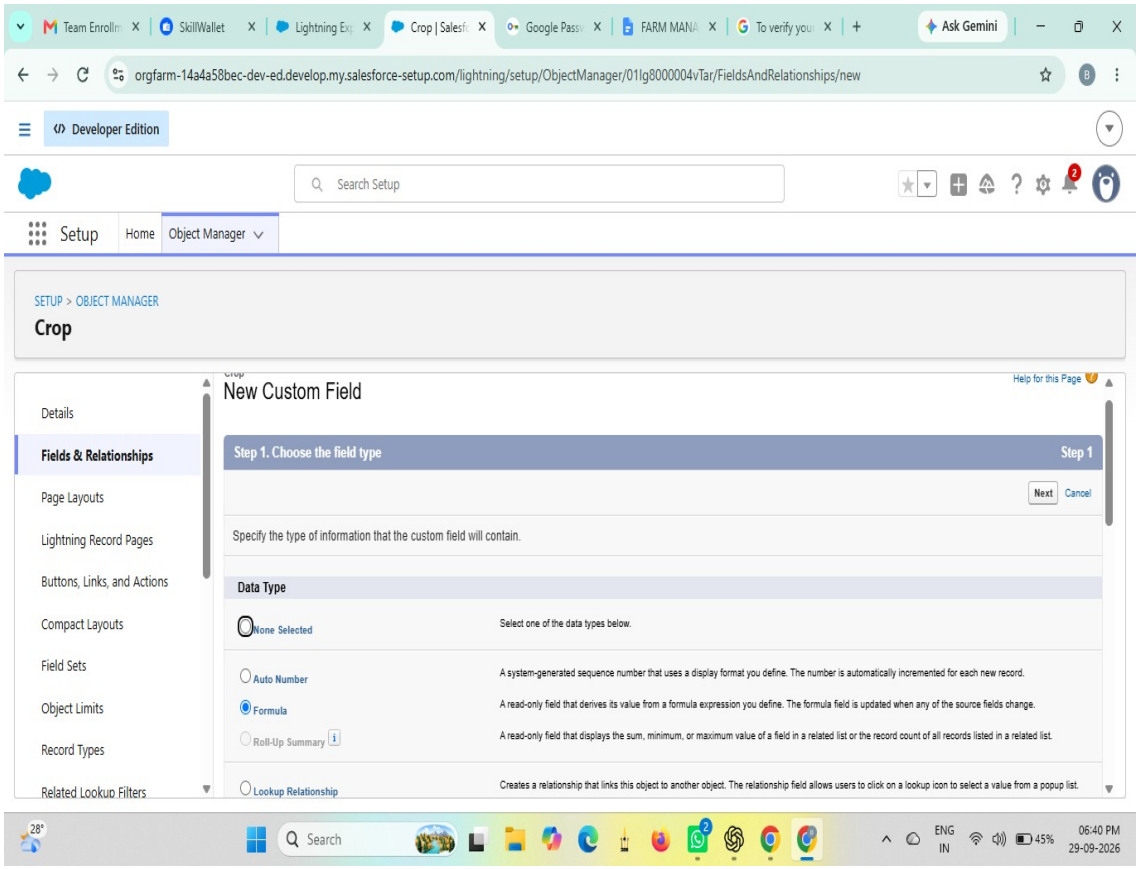
3.Again we select data type as a date,Enter field labe (Harvest date),save

4.Next We Create Lookup relationship on crop object realted to Farm Object.

5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.

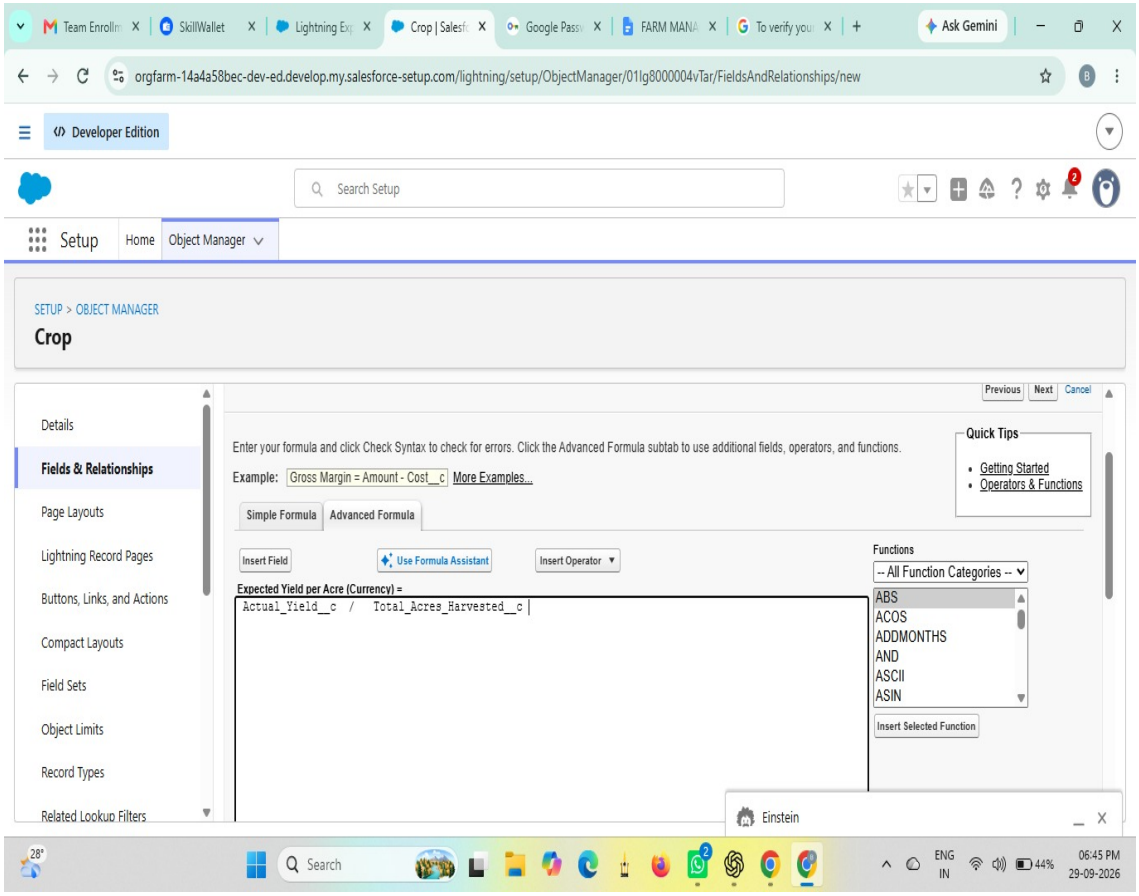
6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.

7.Now We create Formula Field For That Select data Type As a Formula,Click Next.



8.Enter Field Label (Expected Yield per Acre),Select formula return type as Currency,click Next.

9.Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3:

Creation of fields for the Farmer objects:

1.Select Data type Phone,Enter field label(Contact Number),next Save.

2.Select Data type Picklist,Enter Field Label(State),Enter values (Telangana,Andhra Pradesh,Tamilnadu),save.

3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save.

4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of villages,next save.

Creation of field dependency :

1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.

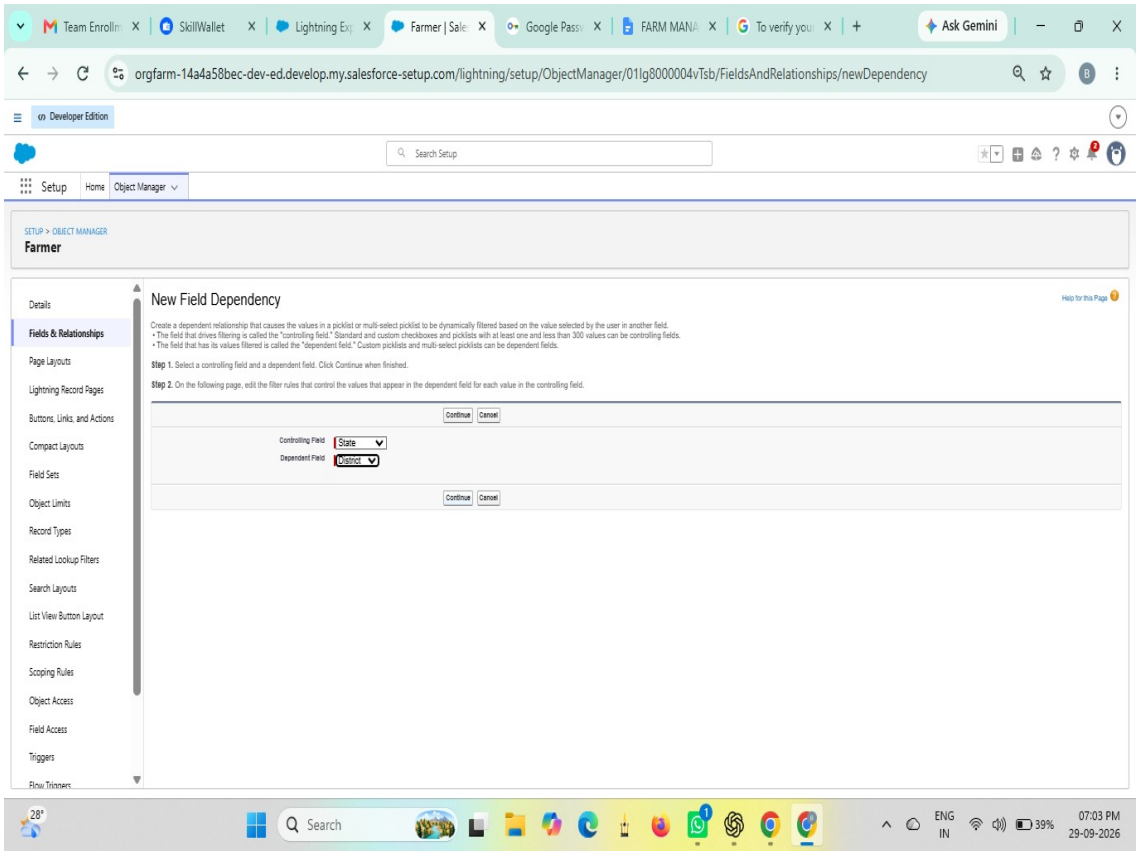
2.Click new, Select Controlling Field as State,Dependent Field As District,Click on Continue.

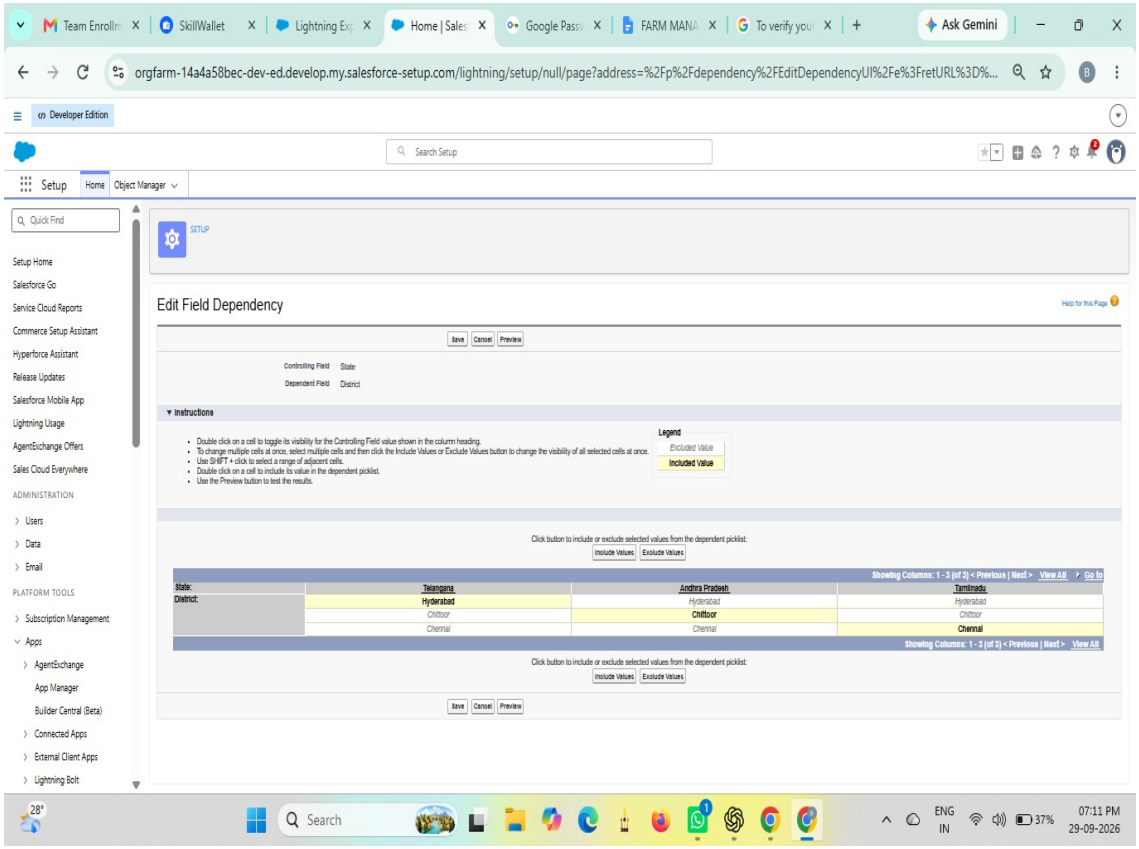
3. Select the appropriate Value in each column by double-clicking them.

4. For Ex. Telangana - Yadadri Bhongir.

5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6.Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.





7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1.Select Data type as a Phone,Enter a Field Label(Contact),Save.

2.Select Data type as an Email,Enter Field Label(Email), Next save.

3.Select Data type Text area,Enter Field Label(Address),Save it.

4.Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.

Milestone 6- Profile:

What is a profile?

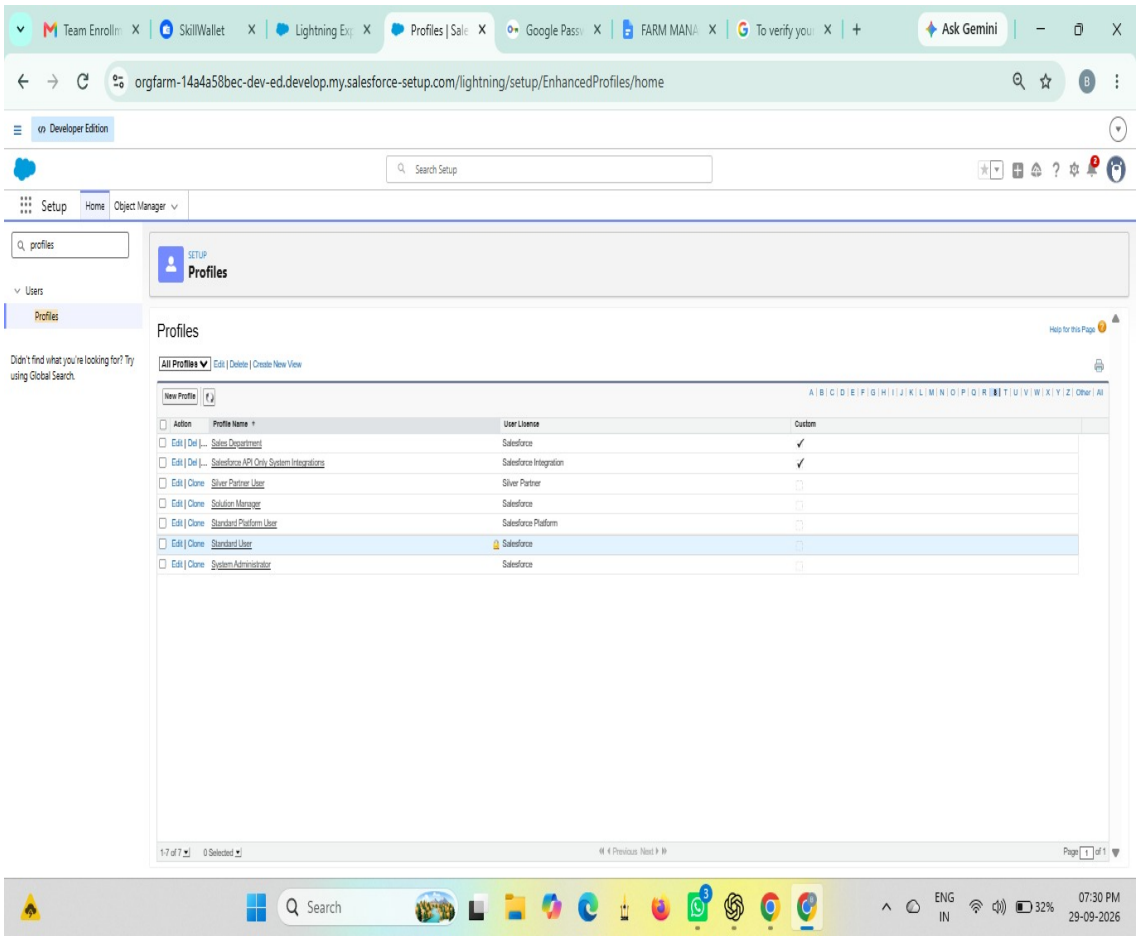
A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record

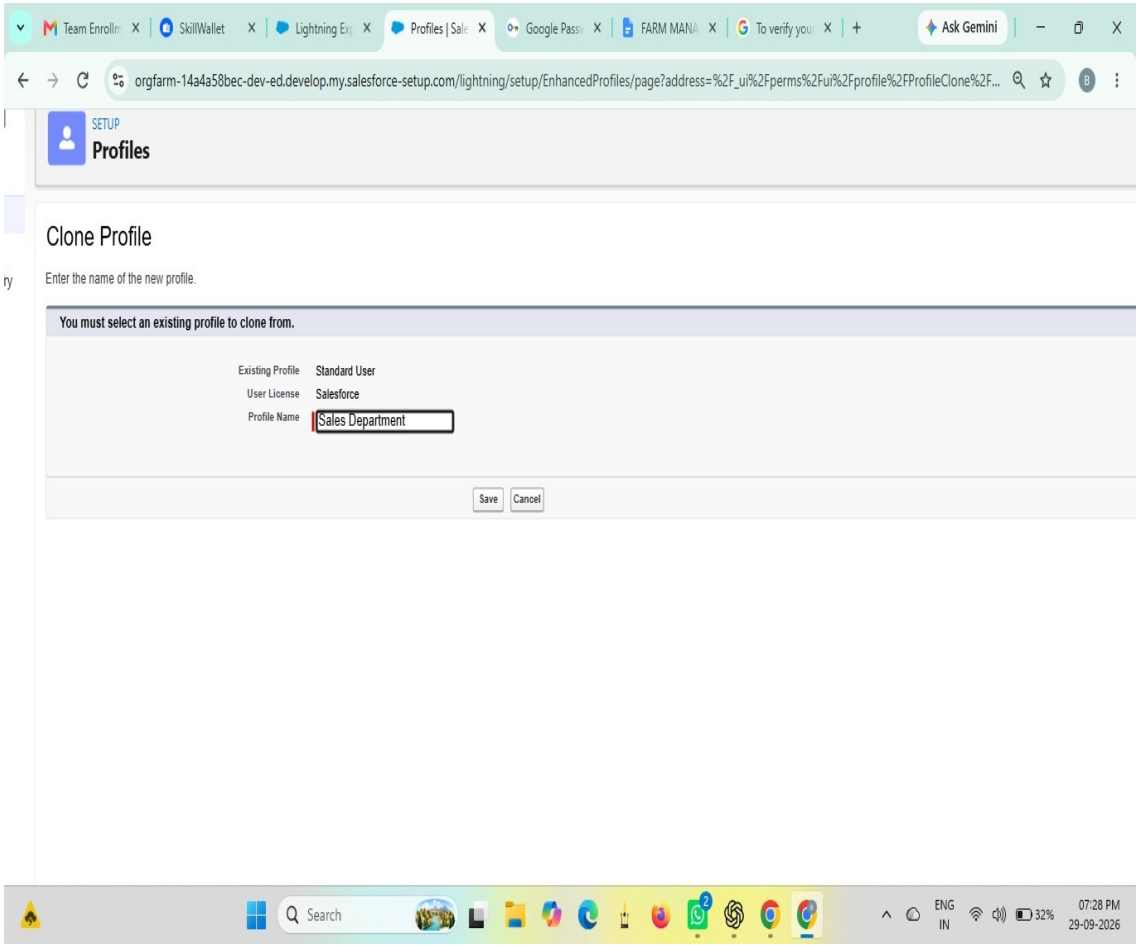
Types, Login hours & Login IP ranges. A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.





5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as " Farm Manager" .

6. From setup , enter profiles in Quick Find box

7. Select profiles (Standard user).

8. Click clone.

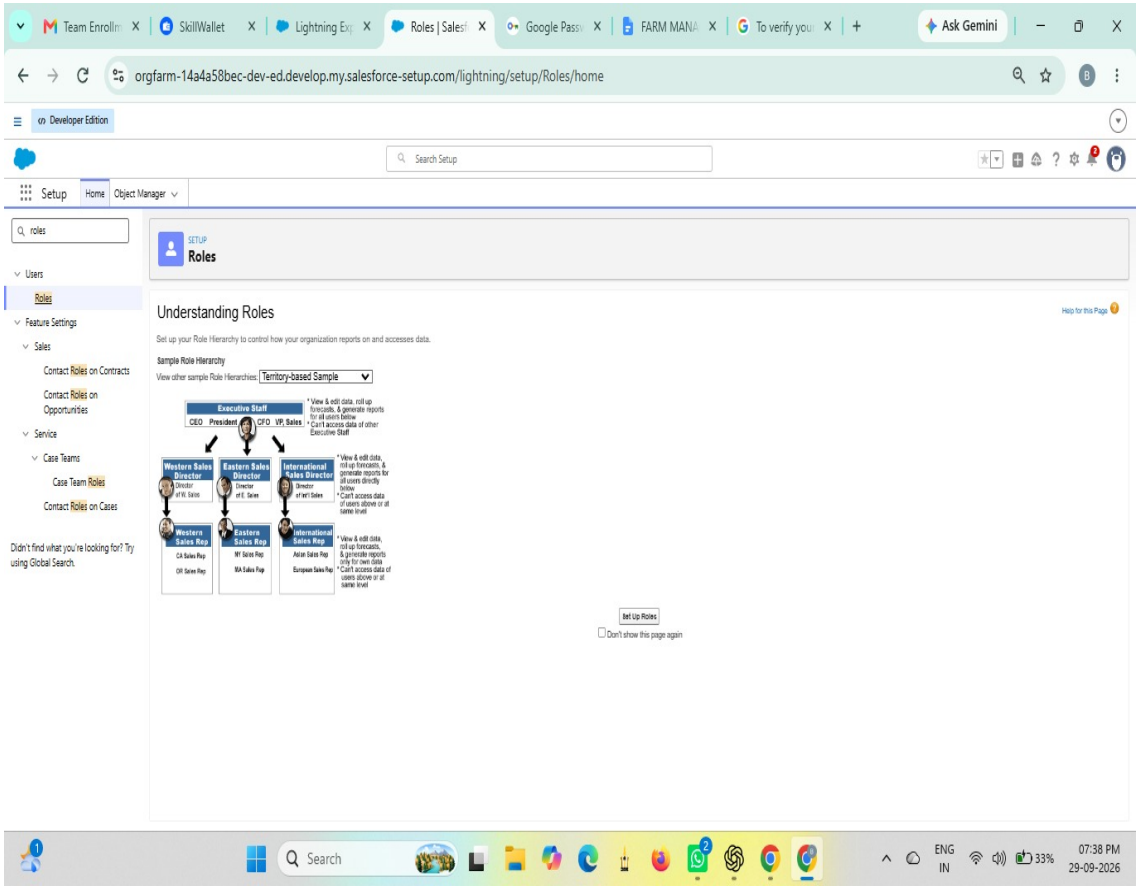
Milestone 7-Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

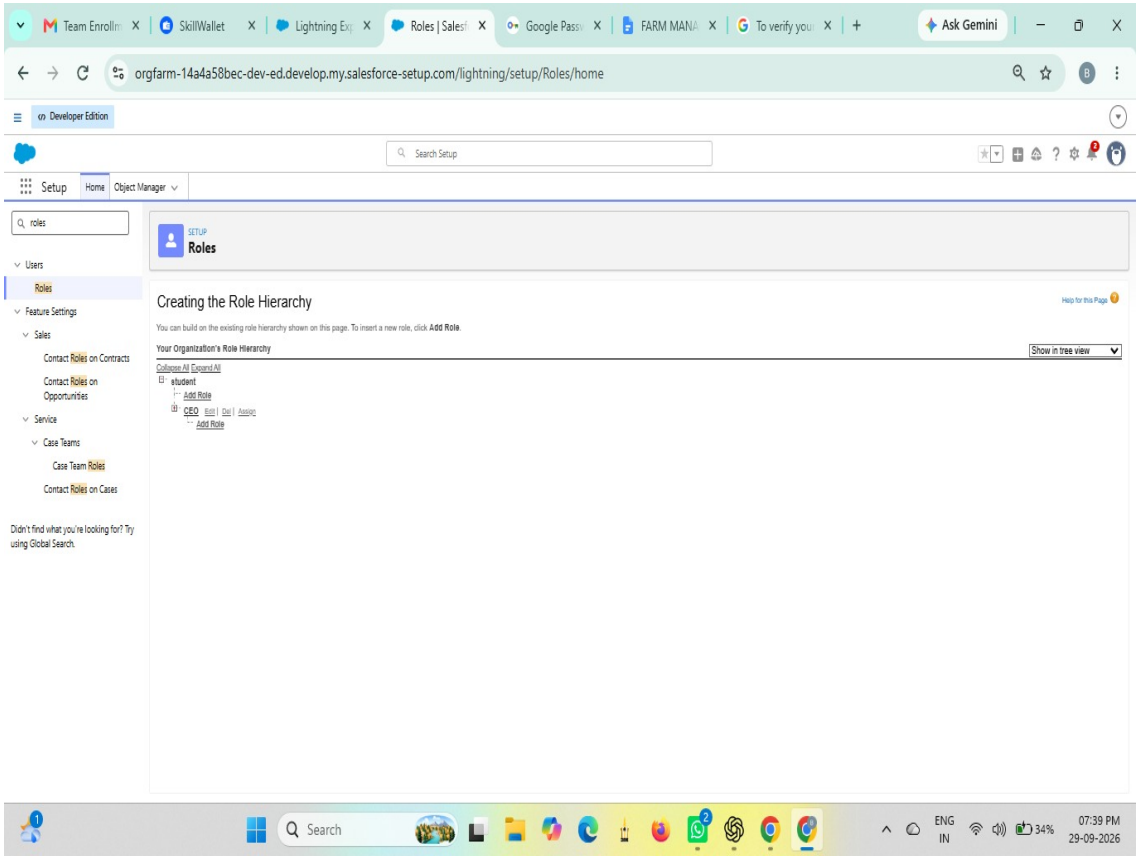
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3.Below the CEO click on add role and enter the label name as a ” Farm Manager ” and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

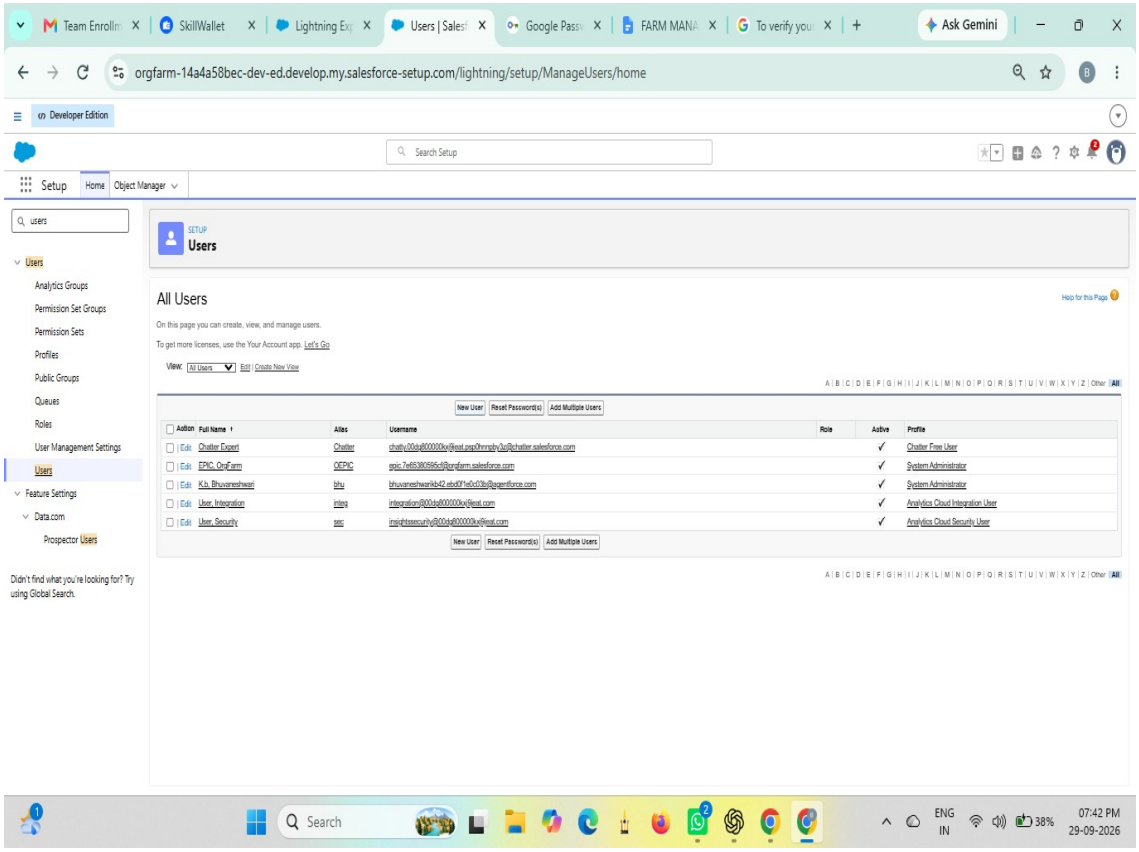
A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don’t find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.

4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.

5. Select the role Farm Manager.

6. Select user License as Standard Platform User.

7. Select profile(Farm Manager).

8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

1) Goto Sharing Settings, which can be found under the Quick Find section.

2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.

3) Add the label of the sharing rule you want to make.

4) Select your rule type based on the criteria.

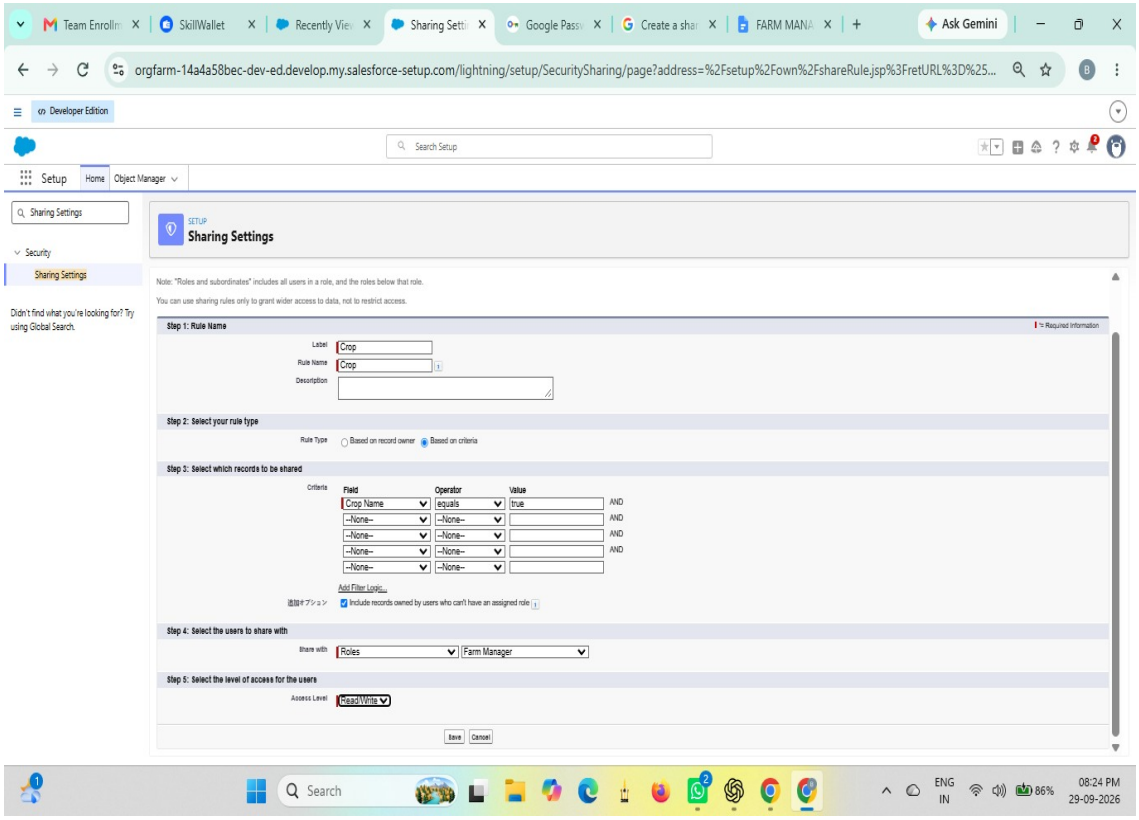
5) Select the field can join immediately check field from the Crop object.

6) Select the Crop as equal and value is True.

7) And in selecting the users to share with the section select roles and in that select Farm Manager.

8) And in the section of select the level of access for the users give the access Read/Write.

9) And save the rule.



Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

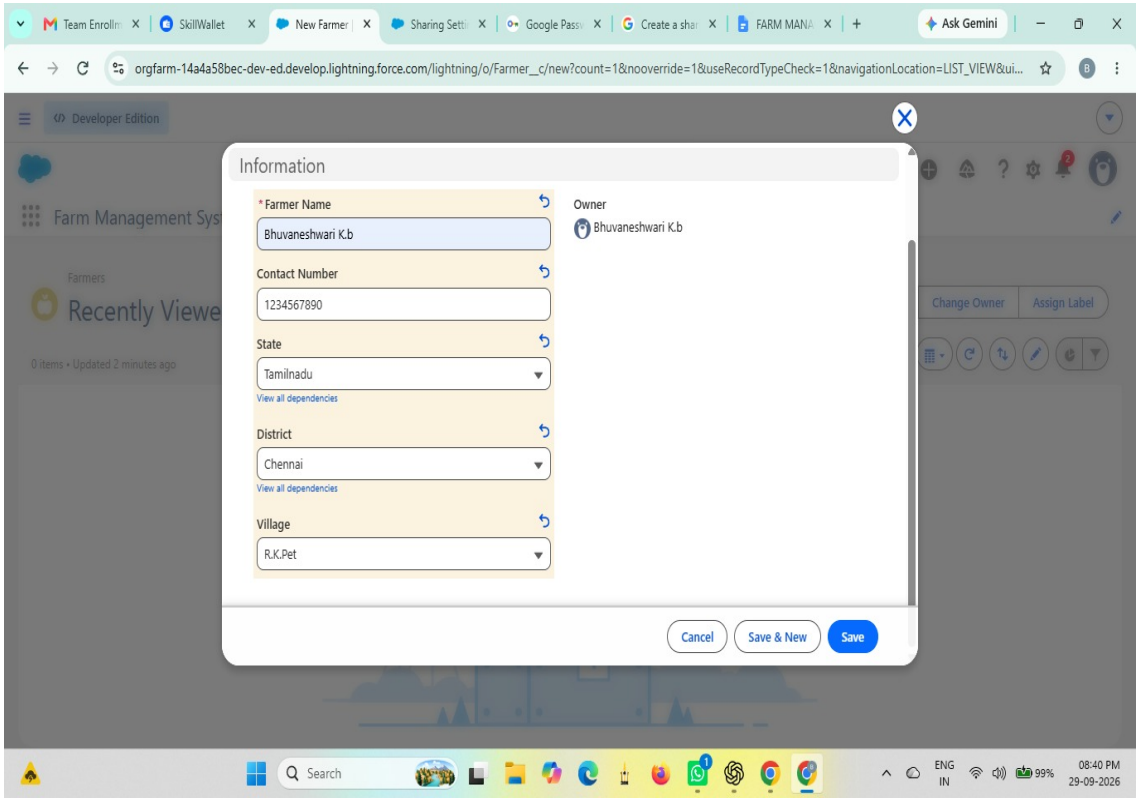
Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1:

Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



Milestone 11-Reports:

What are Reports?

A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

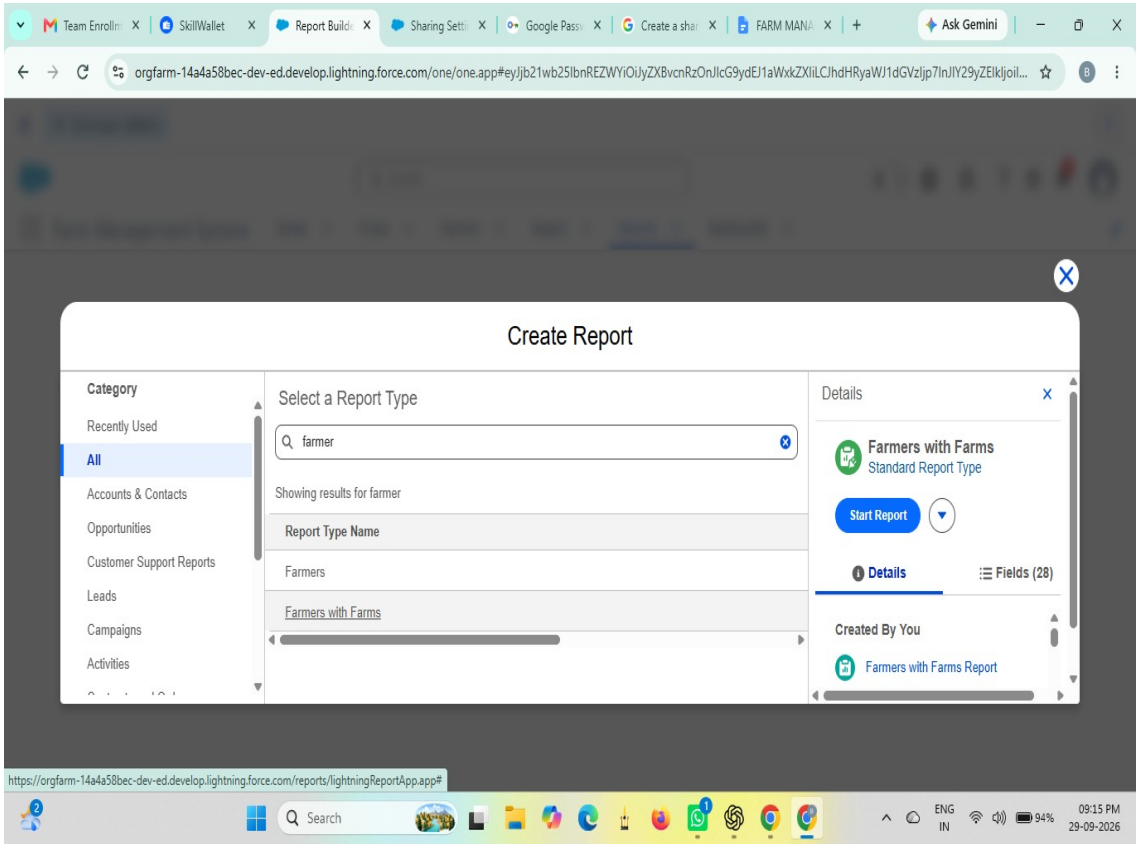
Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

1) Click on app launcher search for reports.

2) Click on the new report and select the category has Farmer with Farms.

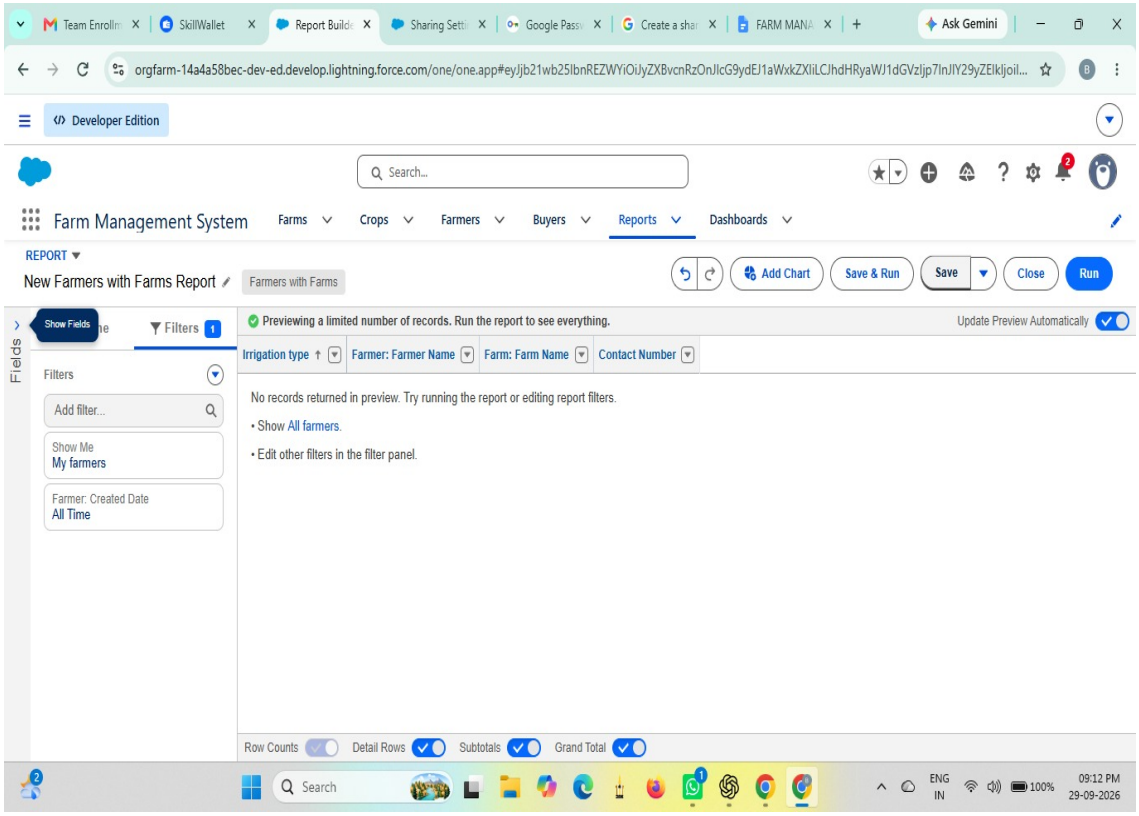


3) In the details section select the option start report.

4) In Outline Click on Column Select Which Field you want to display in the Report.

5) Click Group,Select Which Fields You Want to Group.

6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Milestone 12-Dashboard:

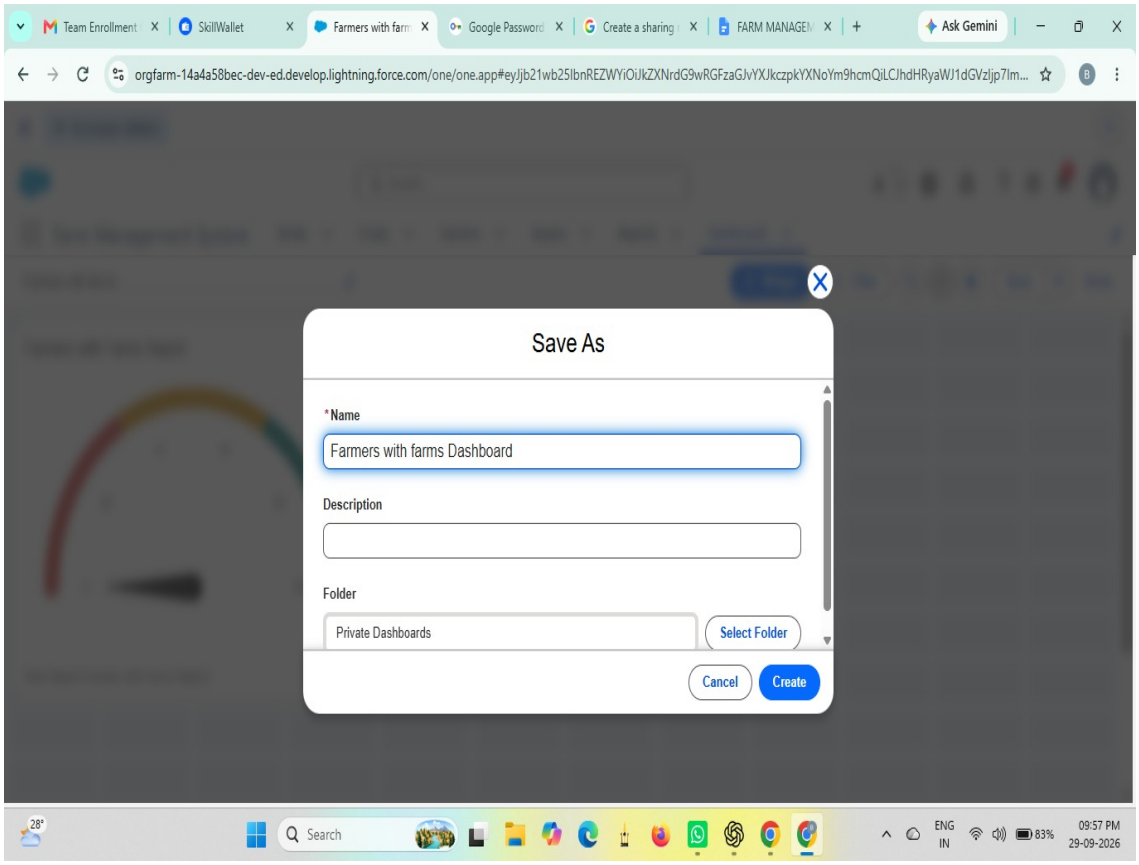
Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.

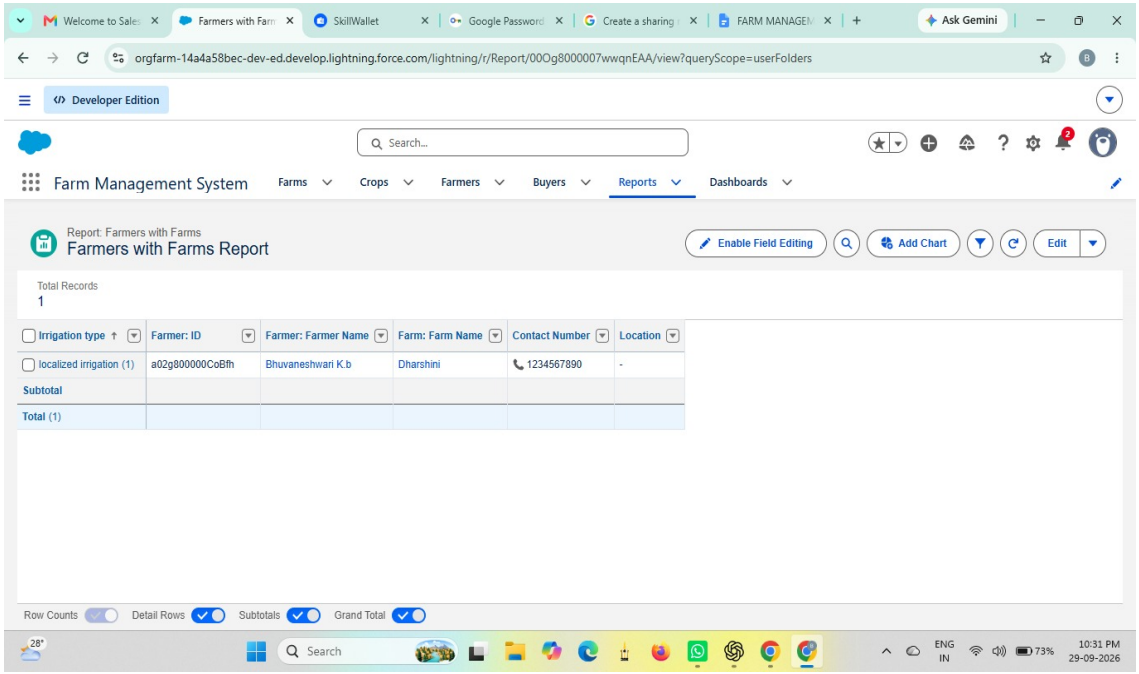
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



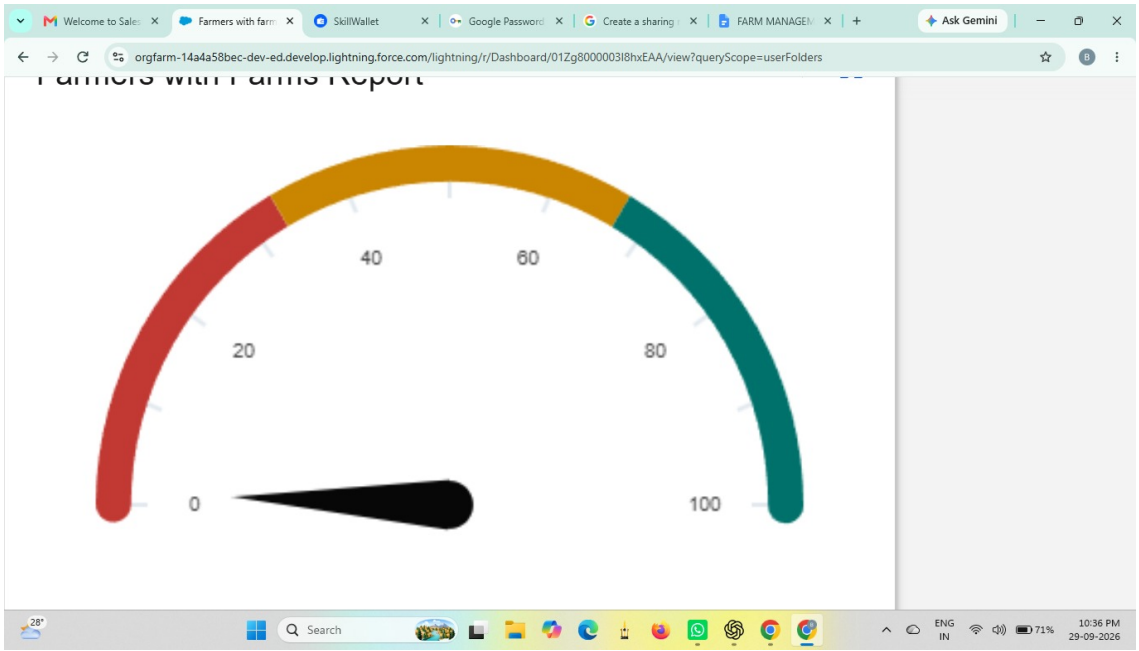
Activity 2: Dashboard

• Click on App Launcher on left side of screen.

• Search Farm Management system & click on it.

• Click on Dashboard Tab.

• Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

The Farm Management System was tested systematically to verify that all the implemented features work according to the specified requirements. Functional testing was performed on the major modules of the application to ensure that users can create, view, update, and manage farm-related information correctly.

The testing process included verification of different Salesforce objects, fields, relationships, validation rules, user permissions, reports, and dashboards. Sample records were entered into the system to check whether the application accepts valid information and prevents incorrect or incomplete data.

The following functionalities were tested:

- Creation and modification of **Farm records**.
- Addition and updating of **Crop information**.
- Creation and maintenance of **Farmer records**.
- Registration and management of **Buyer details**.
- Verification of required fields and input values.
- Testing of validation rules for incorrect or incomplete data.
- Checking relationships between Farm, Crop, Farmer, and Buyer records.
- Verification of user profiles, roles, and permissions.
- Testing of access restrictions for different users.
- Verification of reports generated from stored records.
- Testing of dashboard components and displayed information.
- Checking whether updated records are reflected correctly in reports and dashboards.

Performance Testing

Performance testing was conducted to check the response and stability of the application during regular operations. Activities such as opening records, entering data, updating information, generating reports, and viewing dashboards were performed to observe system responsiveness.

The system was also checked for consistent performance during repeated user operations. The testing helped identify data-entry issues, configuration errors, and access-related problems. Necessary corrections were made before completing the final implementation.

The testing process confirmed that the major features of the Farm Management System function properly and that the application provides a stable environment for managing farm-related information.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

The Farm Management System provides several advantages by replacing traditional record-keeping methods with a centralized digital platform.

Centralized Data Management

All important farm-related information can be maintained in a single Salesforce environment. This makes it easier to store, access, and update records.

Reduced Manual Work

The system reduces the need for maintaining paper registers and separate spreadsheets. Digital records make regular data management more convenient.

Improved Data Accuracy

Validation rules and structured fields help reduce incorrect or incomplete information during data entry.

Easy Monitoring

Farm managers can monitor crop information, farm activities, production, and sales through reports and dashboards.

Better Reporting

The system can generate reports based on stored information, allowing users to review and analyze farm-related data.

Secure Access

User roles, profiles, and permissions can be configured to control access to sensitive information.

Better Decision Support

Reports and dashboards provide organized information that can help users understand farm performance and plan activities more effectively.

Cloud Accessibility

Since the system is implemented using Salesforce, authorized users can access the application through a cloud-based environment.

8.2 Disadvantages

Although the system provides several benefits, it also has some limitations.

Internet Dependency

The application generally requires an internet connection to access the Salesforce cloud environment.

Salesforce Dependency

The functionality of the application depends on the Salesforce platform and its available features.

Limited Offline Usage

Users may have limited access to certain functions when there is no internet connectivity.

Technical Knowledge

Some configuration and maintenance activities may require knowledge of Salesforce administration and development.

Cost Considerations

Depending on the number of users and required Salesforce features, additional licenses or resources may be required for large-scale implementation.

9. CONCLUSION

The **Farm Management System (FMS)** was developed to provide a centralized and organized solution for managing agricultural information. The system addresses common difficulties associated with manual record maintenance, scattered information, data-entry errors, and delayed access to important farm records.

The application allows users to maintain information related to farms, crops, farmers, buyers, production activities, and sales in a structured manner. By using Salesforce features such as custom objects, fields, relationships, validation rules, user permissions, reports, and dashboards, the system provides an integrated environment for managing farm operations.

The implementation of validation rules helps improve the quality and consistency of stored information. Relationships between different records make it easier to connect related farm and agricultural data. Reports and dashboards provide useful summaries that can help users monitor activities and understand important trends.

The system also reduces dependence on paper-based records and repetitive manual processes. Farm managers can access relevant information more efficiently and use the available reports to support operational planning.

Overall, the project demonstrates the practical use of a cloud-based Salesforce platform for agricultural data management. The Farm Management System provides a foundation that can be further improved with advanced technologies such as Artificial Intelligence, IoT, mobile applications, weather services, and predictive analytics.

10. FUTURE SCOPE

The Farm Management System can be enhanced in the future by integrating modern technologies and additional agricultural services.

The system can be integrated with weather services to provide information about rainfall, temperature, humidity, and weather conditions. IoT devices and sensors can be connected to collect real-time information such as soil moisture and temperature, which can help in monitoring farm conditions more effectively. Automated notifications can also be introduced to remind users about important farming activities, crop updates, and sales transactions.

Furthermore, market price information can be integrated to help farmers monitor agricultural product prices and plan their sales. Government scheme and subsidy information can also be included to provide useful details about available agricultural support programs. In the future, predictive analytics can be used to study historical farm data and identify trends in crop production, sales, and overall farm performance.

These future enhancements can make the Farm Management System more intelligent, connected, and useful for modern agricultural management.

10.1 Mobile Application Integration

A dedicated mobile application can be developed to allow farmers and field workers to enter and update information directly from smartphones while working on farms.

10.2 AI-Based Crop Prediction

Artificial Intelligence can be integrated to analyze historical crop data and provide recommendations related to crop selection, production, and cultivation activities.

10.3 Weather Data Integration

The system can be connected with weather APIs to provide information about rainfall, temperature, humidity, and weather forecasts. This can help users plan farming activities more effectively.

10.4 IoT Sensor Connectivity

IoT sensors can be used to collect real-time information such as soil moisture, temperature, humidity, and water levels. This information can be connected to the system for continuous monitoring.

10.5 Market Price Forecasting

Future versions of the system can include agricultural market information and price analysis. This can help farmers understand market trends and plan their sales activities.

10.6 Government Scheme Integration

The application can provide information about agricultural subsidies, financial assistance, crop insurance, and other government-supported farming schemes.

10.7 Automated Notifications

Notifications can be introduced for important activities such as crop maintenance, irrigation schedules, buyer requests, sales updates, and other farm operations.

10.8 Predictive Analytics

Advanced analytics can be used to analyze historical farm data and identify patterns related to crop production, sales, and farm performance.

10.9 Online Buyer Integration

The system can be extended to connect farmers directly with buyers, allowing product availability and transaction information to be managed through the platform.

11. APPENDIX

The appendix contains supporting information and resources related to the development, implementation, and testing of the Farm Management System.

11.1 Salesforce Configuration

The project configuration includes custom objects, fields, relationships, validation rules, profiles, roles, permissions, Lightning application settings, and other Salesforce configurations used to build the system.

11.2 Project Dataset

Sample data was used for development and testing purposes. The dataset includes records related to farms, crops, farmers, buyers, production information, and sales transactions.

11.3 Testing Screenshots

Screenshots captured during the implementation and testing process can be included as evidence of the completed system. These screenshots may show record creation, validation, user access, reports, dashboards, and other important functionalities.

11.4 Reports and Dashboards

The project includes reports and dashboards created to present farm-related information in an organized manner. These provide summaries of crop activities, production, sales, and other important information.

11.5 Tools and Technologies Used

The major tools and technologies used in the project are:

- Salesforce Developer Environment
- Salesforce Lightning Experience
- Custom Objects and Fields
- Validation Rules
- Salesforce Flow

- Reports and Dashboards
- Profiles and Roles
- Permission Sets
- Salesforce Cloud Platform

11.6 Supporting Documentation

The supporting documentation includes the project requirements, customer journey map, empathy analysis, brainstorming activities, solution design, project planning, sprint schedule, testing activities, and implementation details.

These materials provide additional information about the development process and demonstrate how the Farm Management System was planned, designed, implemented, and tested.

THANK YOU